

Welcome to the "Programming: Programming in C" chapter of your GATE Computer Science exam preparation. This section is designed to be a comprehensive, exam-focused reference, providing a quick yet thorough review of all essential concepts, formulas, and problem-solving techniques relevant to C programming for the GATE exam. Mastering C is fundamental not just for direct programming questions but also for understanding underlying principles of Data Structures, Algorithms, Operating Systems, and Computer Architecture. It forms the bedrock of many advanced computer science concepts.

Subject Overview

The "Programming in C" subject is a cornerstone of the GATE Computer Science syllabus, focusing on the syntax, semantics, and practical application of the C programming language. It is crucial for building a strong foundation in computer science, as C's low-level memory management capabilities and efficiency make it indispensable for system programming, embedded systems, and high-performance computing. For GATE CS, this subject typically carries a weightage of 5-10 marks, often integrated with questions from Data Structures and Algorithms. Questions usually involve code snippets for output prediction, error identification, understanding memory allocation, pointer arithmetic, function calls, recursion, and conceptual questions on data types, control flow, and storage classes. A deep understanding of C helps in visualizing program execution and memory layout, which is vital for solving complex problems.

Topic-wise Key Concepts

Programming In C

Definition and Core Idea: C is a general-purpose, procedural, imperative computer programming language developed in the early 1970s by Dennis Ritchie at Bell Labs. It is known for its efficiency, low-level memory access, and portability, making it a popular choice for system programming and embedded systems.

- **Key Properties and Identities:**

- Mid-level language: Combines features of high-level languages with the ability to manipulate memory directly.
- Procedural: Programs are structured as a sequence of function calls.
- Compiled: Source code is translated into machine code by a compiler.
- Strongly typed (but with implicit conversions): Variables have specific types, but C allows flexible type casting.
- Memory management: Manual memory allocation and deallocation using functions like `malloc()` and `free()`.

- **Common Pitfalls or Tricky Points:**

- Undefined behavior: Operations like accessing out-of-bounds array elements or dereferencing a dangling pointer can lead to unpredictable results.
- Memory leaks: Forgetting to `free()` dynamically allocated memory.
- Buffer overflows: Writing past the end of an allocated buffer.

- **Standard Problem-Solving Techniques or Shortcuts:**

- Understand the compilation process (preprocessor, compiler, assembler, linker).
- Always initialize variables to avoid garbage values.
- Pay attention to operator precedence and associativity.

Programming Constructs

Definition and Core Idea: Programming constructs are the fundamental building blocks that control the flow of execution in a program. They include sequential execution, selection (conditional statements), and iteration (loops).

- **Key Properties and Identities:**

- **Sequential:** Statements execute one after another.
- **Selection:**
 - if-else: Executes a block of code based on a condition.


```
if (condition) { // code } else { // code }
```
 - switch-case: Multi-way branching based on the value of an integer expression.
- **Iteration (Loops):**
 - for loop: Used when the number of iterations is known.

```
for (initialization; condition; increment/decrement) { // code }
```

- while loop: Executes as long as a condition is true (entry-controlled).

```
while (condition) { // code }
```

- do-while loop: Executes at least once, then continues as long as a condition is true (exit-controlled).

```
do { // code } while (condition);
```

- **Jump Statements:** break, continue, goto, return.
- **Common Pitfalls or Tricky Points:**
 - Off-by-one errors in loops.
 - Infinite loops due to incorrect loop conditions or missing increment/decrement.
 - Forgetting break in switch-case, leading to fall-through.
 - Using = (assignment) instead of == (comparison) in conditions.
- **Standard Problem-Solving Techniques or Shortcuts:**
 - Trace the execution path of the program step-by-step for conditional and loop statements.
 - For loops, identify the initial state, the termination condition, and how variables change in each iteration.

Programming Paradigms

Definition and Core Idea: A programming paradigm is a fundamental style of computer programming, providing a way to classify programming languages based on their features. C primarily adheres to the procedural paradigm.

- **Key Properties and Identities:**
 - **Procedural Programming:** Focuses on a sequence of instructions (procedures or functions) to perform computations. Data and functions are separate. C, Fortran, Pascal are examples.
 - **Imperative Programming:** Programs explicitly state how the computation should be performed, detailing changes to the program state. C is an imperative language.
 - Contrast with Object-Oriented (data and methods encapsulated) and Functional (computation as evaluation of mathematical functions, avoiding state changes) paradigms.
- **Common Pitfalls or Tricky Points:**
 - Confusing C's procedural nature with object-oriented concepts like classes and objects, which are not directly supported.
 - Trying to apply functional programming principles (like immutability) where C's mutable state is fundamental.
- **Standard Problem-Solving Techniques or Shortcuts:**
 - Understand that C programs are executed as a series of instructions that modify program state.
 - Focus on function calls, parameter passing, and global/local variable interactions.

Variable Binding

Definition and Core Idea: Variable binding refers to the process of associating attributes (like type, value, and storage location) with a variable name. This binding can occur at different stages of a program's lifecycle.

- **Key Properties and Identities:**
 - **Static Binding (Compile-time):** Attributes are determined before runtime and remain fixed. E.g., type binding for most variables in C.
 - **Dynamic Binding (Run-time):** Attributes are determined during program execution. E.g., value binding, or in some languages, type binding (polymorphism).
 - **Storage Class Binding:** Determines the scope and lifetime of a variable.
 - auto: Local to a block, created on entry, destroyed on exit (default for local variables).
 - static: Local to a block/file, created once, persists throughout program execution.
 - extern: Global variable, declared in one file, defined in another.
 - register: Suggests storing variable in CPU register for faster access (compiler may ignore).
 - **Scope:** Region of code where a variable is accessible (block scope, function scope, file scope, program scope).
 - **Lifetime:** Period during which a variable exists in memory.
- **Common Pitfalls or Tricky Points:**
 - Confusing scope and lifetime: A variable might be alive but out of scope.
 - Modifying static local variables across multiple function calls.
 - Shadowing: A local variable having the same name as a global variable.
- **Standard Problem-Solving Techniques or Shortcuts:**
 - Draw scope boxes for functions and blocks to visualize variable accessibility.
 - Track variable values across function calls, especially for static variables.

Type Checking

Definition and Core Idea: Type checking is the process of verifying and enforcing the type constraints of a programming language. It ensures that operations are applied to compatible data types, preventing common programming errors.

- **Key Properties and Identities:**

- **Static Type Checking (Compile-time):** Types are checked before program execution. C is primarily statically typed.
- **Dynamic Type Checking (Run-time):** Types are checked during program execution.
- **Type Compatibility:** Rules for when types can be used together (e.g., implicit conversions).
- **Type Casting:** Explicit conversion of one data type to another using (type)expression.
- **Implicit Type Promotion (Integral Promotion):** Smaller integer types (char, short) are promoted to int in expressions.
- **Arithmetic Conversion:** In binary operations, operands are converted to a common type (usually the "larger" or "wider" type).

- **Common Pitfalls or Tricky Points:**

- Unexpected implicit type conversions, especially between signed and unsigned integers, or between integers and floating-point types, leading to loss of precision or incorrect results.
- Incorrect type casting, e.g., casting a pointer to an incompatible type and then dereferencing it.
- Integer division truncates the fractional part (e.g., $5/2 = 2$).

- **Standard Problem-Solving Techniques or Shortcuts:**

- Always be aware of the data types involved in an expression.
- Use explicit type casting when precision or specific type behavior is required.
- Remember that sizeof operator returns an unsigned integer type (size_t).

Output

Definition and Core Idea: Output refers to the process of displaying information from a program to the user or writing it to a file. In C, standard output functions like printf(), puts(), and putchar() are commonly used.

- **Key Properties and Identities:**

- printf(const char *format, ...): Formatted output to standard output.
 - Format specifiers:
 - %d or %i: signed decimal integer
 - %u: unsigned decimal integer
 - %f: decimal floating point (double by default)
 - %c: character
 - %s: string
 - %p: pointer address (hexadecimal)
 - %x or %X: hexadecimal integer
 - %o: octal integer
 - %%%: print a literal % character
 - Modifiers: .precision, width, - (left-align), + (sign), 0 (zero-pad), l (long), ll (long long), h (short).
- puts(const char *str): Writes a string to standard output, followed by a newline.
- putchar(int char_val): Writes a single character to standard output.

- **Common Pitfalls or Tricky Points:**

- Mismatched format specifiers and argument types can lead to undefined behavior or incorrect output.
- Forgetting newline characters \n in printf() (puts() adds it automatically).
- Buffer flushing issues, especially when mixing buffered and unbuffered I/O.

- **Standard Problem-Solving Techniques or Shortcuts:**

- Carefully match each format specifier in printf() with its corresponding argument.
- Trace the exact values and types of variables being printed.
- Understand the effect of width and precision specifiers on output formatting.

Array

Definition and Core Idea: An array is a collection of elements of the same data type, stored in contiguous memory locations. Elements are accessed using an index, typically starting from zero.

- **Key Properties and Identities:**

- Contiguous memory allocation.

- Fixed size at compile time (for static arrays).
- Zero-indexed: The first element is at index 0.
- Array name decays to a pointer to its first element in most contexts.
- **Address Calculation (1D Array):**

$$\text{Address}(A[i]) = \text{BaseAddress} + i \times \text{sizeof}(\text{ElementType})$$

- **Address Calculation (2D Array, Row-Major):**

$$\text{Address}(A[i][j]) = \text{BaseAddress} + (i \times \text{cols} + j) \times \text{sizeof}(\text{ElementType})$$

where cols is the number of columns in the array.

- **Common Pitfalls or Tricky Points:**

- Array out-of-bounds access: Accessing elements beyond the declared size leads to undefined behavior.
- When an array is passed to a function, it decays into a pointer, losing its size information.
sizeof(array_in_function) will return the size of a pointer, not the array.
- Multidimensional arrays are stored in row-major order in C.

- **Standard Problem-Solving Techniques or Shortcuts:**

- Draw memory diagrams to visualize array elements and their addresses.
- Always check loop bounds to prevent out-of-bounds access.
- Remember that array indexing $A[i]$ is equivalent to pointer arithmetic $*(A + i)$.

Strings

Definition and Core Idea: In C, a string is a sequence of characters stored in a character array, terminated by a null character (`\0`). This null terminator marks the end of the string.

- **Key Properties and Identities:**

- Declared as `char array_name[]` or `char *pointer_name`.
- Always null-terminated. The size of the array must be at least one greater than the number of characters in the string to accommodate `\0`.
- Standard library functions (from `<string.h>`):
 - `strlen(const char *s)`: Returns the length of the string (excluding `\0`).
 - `strcpy(char *dest, const char *src)`: Copies `src` to `dest`.
 - `strncpy(char *dest, const char *src, size_t n)`: Copies at most `n` characters. Does not guarantee null termination if `src` is longer than `n`.
 - `strcat(char *dest, const char *src)`: Appends `src` to `dest`.
 - `strncat(char *dest, const char *src, size_t n)`: Appends at most `n` characters.
 - `strcmp(const char *s1, const char *s2)`: Compares `s1` and `s2` lexicographically. Returns 0 if equal, `<0` if `s1 < s2`, `>0` if `s1 > s2`.
 - `strncmp(const char *s1, const char *s2, size_t n)`: Compares at most `n` characters.

- **Common Pitfalls or Tricky Points:**

- Buffer overflow: Using `strcpy()` or `strcat()` without ensuring the destination buffer is large enough. Always prefer `strncpy()` and `strncat()` with careful handling of null termination.
- Forgetting the null terminator `\0`, leading to functions like `strlen()` reading past the allocated memory.
- Modifying string literals (e.g., `char *s = "hello"; s[0] = 'H';`) leads to undefined behavior, as string literals are often stored in read-only memory.

- **Standard Problem-Solving Techniques or Shortcuts:**

- Always allocate sufficient memory for strings, considering the null terminator.
- When dealing with string manipulation, mentally trace the contents of the character arrays, including the `\0`.
- Be careful with pointer arithmetic on string pointers.

Pointers

Definition and Core Idea: A pointer is a variable that stores the memory address of another variable. It allows for indirect access to data and is fundamental for dynamic memory management, arrays, and complex data structures in C.

- **Key Properties and Identities:**

- **Declaration:** `type *pointer_name;`
- **Address-of operator:** `&` (returns the memory address of a variable).
- **Dereference operator:** `*` (accesses the value at the address stored in a pointer).
- **Pointer Arithmetic:**
 - Adding an integer to a pointer: `p + n` points to the memory location $n \times \text{sizeof}(*p)$ bytes away from `p`.

$$\text{Address}(p + n) = \text{Address}(p) + n \times \text{sizeof}(*p)$$

- Subtracting an integer from a pointer: $p - n$.
- Subtracting two pointers (of the same type): $p - q$ gives the number of elements between them.

$$p - q = \frac{\text{Address}(p) - \text{Address}(q)}{\text{sizeof}(*p)}$$

- Pointers can be compared for equality or order.
- **void* (Generic Pointer):** Can point to any data type but cannot be dereferenced directly or used in arithmetic without casting.
- **NULL Pointer:** A pointer that points to no valid memory location.
- **Common Pitfalls or Tricky Points:**
 - **Dangling Pointers:** Pointers that point to memory that has been deallocated or is no longer valid.
 - **Wild Pointers:** Uninitialized pointers that point to arbitrary memory locations.
 - Dereferencing a NULL pointer or a wild pointer leads to segmentation faults or undefined behavior.
 - Incorrect pointer arithmetic (e.g., adding incompatible types, or arithmetic on void* without casting).
 - Confusing *p (value at address p) with p (the address itself).
- **Standard Problem-Solving Techniques or Shortcuts:**
 - Draw memory diagrams showing variable names, their addresses, and their values.
 - Mentally trace pointer assignments and dereferencing operations.
 - Always initialize pointers to NULL if they don't point to valid memory immediately.
 - Check for NULL before dereferencing pointers.

Aliasing

Definition and Core Idea: Aliasing occurs when multiple distinct names or pointers refer to the same memory location. In C, this often happens with pointers, array names, or when passing arguments by reference (using pointers).

- **Key Properties and Identities:**
 - If p_1 and p_2 are pointers and $p_1 = p_2$, then both p_1 and p_2 alias the same memory location.
 - Modifying data through one alias will affect the data accessed through other aliases.
 - Can occur with function parameters if pointers are passed.
- **Common Pitfalls or Tricky Points:**
 - Unexpected side effects: Changes made through one alias might inadvertently affect other parts of the program that use a different alias to the same data.
 - Difficult to optimize for compilers due to potential data dependencies.
 - Can lead to subtle bugs that are hard to trace.
- **Standard Problem-Solving Techniques or Shortcuts:**
 - Draw memory diagrams to explicitly show when multiple pointers point to the same location.
 - Be extra cautious when modifying data through pointers, especially when those pointers might be aliases.
 - In GATE questions, always consider if multiple variables/pointers might be referring to the same underlying memory.

Functions

Definition and Core Idea: A function is a self-contained block of code that performs a specific task. Functions promote modularity, reusability, and readability in C programs.

- **Key Properties and Identities:**
 - **Declaration (Prototype):** Specifies the function's return type, name, and parameter types (e.g., `int add(int a, int b);`).
 - **Definition:** Contains the actual code of the function.
 - **Call:** Invokes the function's execution.
 - **Return Type:** The data type of the value the function sends back to the caller. void if no value is returned.
 - **Parameters (Arguments):** Values passed to the function.
 - **Local Variables:** Declared inside a function, have function scope and automatic lifetime.
 - **Global Variables:** Declared outside any function, have file scope and static lifetime.
- **Common Pitfalls or Tricky Points:**
 - Scope of local variables: Local variables cease to exist once the function returns. Returning a pointer to a local variable leads to a dangling pointer.
 - Side effects: Functions modifying global variables or parameters passed by reference (pointers) can lead to

hard-to-track bugs.

- Function prototypes are crucial for correct compilation, especially when functions are defined after their calls.
- **Standard Problem-Solving Techniques or Shortcuts:**
 - Trace function calls using a call stack model, keeping track of local variables and parameters for each active function.
 - Clearly distinguish between parameters (formal arguments) and arguments (actual arguments).
 - Understand how return values are passed back to the caller.

Parameter Passing

Definition and Core Idea: Parameter passing refers to the mechanism by which arguments (actual parameters) are passed from the calling function to the called function (formal parameters).

- **Key Properties and Identities:**

- **Call by Value (C's default):** A copy of the actual argument's value is passed to the formal parameter. Changes to the formal parameter inside the function do not affect the original actual argument in the caller.

```
void func(int x) { x = x + 1; } // x is a copy
```

- **Call by Reference (Simulated in C using Pointers):** The address of the actual argument is passed. The formal parameter is a pointer. Changes made through the pointer inside the function directly modify the original actual argument in the caller.

```
void func(int *ptr) { *ptr = *ptr + 1; } // *ptr modifies original variable
```

- Arrays are always passed by reference (decay to a pointer to their first element).

- **Common Pitfalls or Tricky Points:**

- Misunderstanding when a function can modify the caller's variables. Only possible with call by reference (pointers).
- Forgetting to dereference a pointer when intending to modify the original value in call by reference.
- Passing large structures by value can be inefficient due to copying.

- **Standard Problem-Solving Techniques or Shortcuts:**

- When tracing code, explicitly note whether a variable is being passed by value (copy) or by reference (address).
- For call by reference, always draw an arrow from the pointer parameter to the actual variable it points to in the caller's memory.

Recursion

Definition and Core Idea: Recursion is a programming technique where a function calls itself, either directly or indirectly, to solve a problem. It's often used for problems that can be broken down into smaller, similar subproblems.

- **Key Properties and Identities:**

- **Base Case:** A condition that stops the recursion, preventing an infinite loop. Without a base case, recursion leads to stack overflow.
- **Recursive Step:** The part of the function that calls itself with a modified input, moving closer to the base case.
- Every recursive function can be rewritten iteratively, and vice-versa.
- Often involves a call stack: Each recursive call adds a new frame to the stack.
- **Tail Recursion:** A special form where the recursive call is the last operation in the function. Some compilers can optimize this to iterative code.

- **Formulas and Results (for complexity analysis):**

- Recurrence relations are used to analyze the time and space complexity of recursive algorithms. E.g., for factorial: $T(n) = T(n-1) + O(1)$.
- Space complexity often depends on the maximum depth of the recursion stack.

- **Common Pitfalls or Tricky Points:**

- Missing or incorrect base case, leading to infinite recursion and stack overflow.
- Excessive recursion depth can lead to stack overflow even with a correct base case.
- Difficulty in tracing complex recursive calls.
- Redundant computations in non-optimized recursive solutions (e.g., naive Fibonacci).

- **Standard Problem-Solving Techniques or Shortcuts:**

- Identify the base case first.
- Assume the recursive call works correctly for smaller inputs.
- Trace the execution for small inputs, drawing the call stack to visualize the flow and variable states.
- For GATE, often involves predicting output or determining the number of function calls.

Structure

Definition and Core Idea: A structure (struct) in C is a user-defined data type that groups together variables of different data types under a single name. It allows for creating complex data types that represent real-world entities.

- **Key Properties and Identities:**

- Declared using the struct keyword.
- Members are stored in contiguous memory, but padding may be inserted by the compiler for alignment.
- **Member Access:**
 - Dot operator (.) for structure variables: struct_var.member
 - Arrow operator (->) for pointers to structures: struct_ptr->member (equivalent to (*struct_ptr).member)
- **sizeof(struct):** The size of a structure is at least the sum of the sizes of its members, but due to padding, it can be larger. The compiler aligns members to optimize memory access.

$$\text{sizeof}(\text{struct}) \geq \sum \text{sizeof}(\text{member})$$

- Structures can contain members of other structure types or pointers to themselves (for linked lists, trees).

- **Common Pitfalls or Tricky Points:**

- Understanding memory alignment and padding, which affects sizeof() and memory efficiency.
- Confusing . and -> operators.
- Passing structures by value can be inefficient for large structures; passing by pointer is often preferred.
- Self-referential structures (e.g., for linked lists) require careful handling of pointers.

- **Standard Problem-Solving Techniques or Shortcuts:**

- When calculating sizeof(struct), consider the alignment requirements of each member (usually to its own size or the largest member's size).
- Draw diagrams for structures, especially those with pointers or nested structures, to visualize memory layout.

Union

Definition and Core Idea: A union is a special user-defined data type in C that allows different data types to be stored in the same memory location. Only one member of the union can hold a value at any given time.

- **Key Properties and Identities:**

- Declared using the union keyword.
- All members share the same memory space.
- **sizeof(union):** The size of a union is equal to the size of its largest member, ensuring enough space for any member.

$$\text{sizeof}(\text{union}) = \max(\text{sizeof}(\text{member}_1), \text{sizeof}(\text{member}_2), \dots)$$

- Accessing a member that was not the last one written to results in undefined behavior (unless it's a "type-punning" scenario which is implementation-defined).

- **Common Pitfalls or Tricky Points:**

- Accessing an inactive member (a member that was not the last one assigned a value) leads to undefined behavior.
- Unions are primarily used for memory optimization or type punning (interpreting the same memory in different ways).
- Understanding that changing one member's value overwrites the previous member's value.

- **Standard Problem-Solving Techniques or Shortcuts:**

- When calculating sizeof(union), simply find the largest member's size.
- Mentally track which member of the union is currently "active" (last written to) to predict output.

Switch Case

Definition and Core Idea: The switch-case statement is a multi-way branch control statement that allows a program to execute different blocks of code based on the value of a single expression.

- **Key Properties and Identities:**

- The switch expression must evaluate to an integer type (char, short, int, long, long long, or an enumeration type).
- case labels must be constant integer expressions.
- break statement: Exits the switch block.
- default label: Optional, executed if no case matches.

- **Fall-through:** If a break statement is omitted, execution "falls through" to the next case label.
- **Common Pitfalls or Tricky Points:**
 - Forgetting break statements, leading to unintended fall-through and incorrect logic. This is a very common GATE question trap.
 - Using non-integer expressions or non-constant values for case labels.
 - Placing statements before the first case label (these will always execute).
- **Standard Problem-Solving Techniques or Shortcuts:**
 - Carefully trace the execution path, paying close attention to the presence or absence of break statements.
 - Identify the value of the switch expression and which case (or default) it matches.

Goto

Definition and Core Idea: The goto statement provides an unconditional jump from one point in a function to another labeled point within the same function. It alters the normal sequential flow of control.

- **Key Properties and Identities:**
 - Syntax: goto label; and label: statement;
 - The label must be within the same function.
 - Can be used to break out of nested loops or handle error conditions.
- **Common Pitfalls or Tricky Points:**
 - Generally discouraged in modern programming practice as it can lead to "spaghetti code" that is difficult to read, debug, and maintain.
 - Jumping into a block can bypass variable initialization, leading to undefined behavior.
 - Jumping out of a block can skip destructors (though C doesn't have explicit destructors like C++).
- **Standard Problem-Solving Techniques or Shortcuts:**
 - When encountering goto, simply follow the jump to the specified label to trace the control flow.
 - Understand its direct impact on program execution flow.

Identify Function

Definition and Core Idea: This topic refers to the ability to analyze a given C function's code, its signature (return type, name, parameters), and its interactions with other parts of the program to determine its precise purpose, behavior, and potential side effects.

- **Key Properties and Identities:**
 - **Function Signature:** return_type function_name(parameter_list); provides initial clues about what the function takes and what it returns.
 - **Function Body:** The statements within the function define its logic.
 - **Side Effects:** Changes a function makes to the program state outside its local scope (e.g., modifying global variables, parameters passed by reference, performing I/O).
 - **Pure Functions:** Functions that, given the same input, always return the same output and have no side effects. (Rare in C, but a useful concept).
- **Common Pitfalls or Tricky Points:**
 - Misinterpreting the purpose due to complex logic or subtle side effects.
 - Not recognizing the impact of parameter passing mechanisms (call by value vs. call by reference).
 - Overlooking hidden dependencies on global variables or external resources.
- **Standard Problem-Solving Techniques or Shortcuts:**
 - Trace the function with a few sample inputs, noting the return value and any changes to external variables.
 - Break down complex functions into smaller, understandable parts.
 - Look for common patterns: mathematical operations, array/string manipulations, recursive calls, pointer operations.

Loop Invariants

Definition and Core Idea: A loop invariant is a condition or property that holds true before the first iteration of a loop, remains true before each subsequent iteration, and is true after the loop terminates. It is a powerful tool for proving the correctness of loops and algorithms.

- **Key Properties and Identities:**
 - **Initialization:** The invariant must be true before the first iteration of the loop.
 - **Maintenance:** If the invariant is true before an iteration, it must remain true after that iteration (before the next).
 - **Termination:** When the loop terminates, the invariant, combined with the loop termination condition, should

imply the desired property of the algorithm.

- **Common Pitfalls or Tricky Points:**

- Identifying the correct loop invariant for a given problem.
- Rigorously proving all three properties (initialization, maintenance, termination).
- Confusing the loop invariant with the loop condition or the desired post-condition.

- **Standard Problem-Solving Techniques or Shortcuts:**

- For common algorithms (e.g., sorting, searching), try to recall standard loop invariants.
- To find an invariant, consider what property is preserved or incrementally built up by each iteration towards the final solution.
- Useful for understanding why an algorithm works, not just how.

Runtime Environment

Definition and Core Idea: The runtime environment refers to the state of a program during its execution, including its memory layout, the call stack, heap, and interaction with the operating system. Understanding this is crucial for debugging and optimizing C programs.

- **Key Properties and Identities:**

- **Memory Layout:**

- **Text Segment:** Stores compiled code (read-only).
- **Data Segment:** Stores global and static variables (initialized data).
- **BSS Segment:** Stores uninitialized global and static variables (zero-initialized by OS).
- **Heap:** Dynamically allocated memory (malloc(), calloc(), realloc(), free()). Grows upwards.
- **Stack:** Stores local variables, function parameters, and return addresses for function calls. Grows downwards.

- **Call Stack:** A stack data structure that stores information about the active subroutines (functions) of a computer program. Each function call creates a "stack frame."

- **Common Pitfalls or Tricky Points:**

- **Stack Overflow:** Occurs when the call stack runs out of memory, often due to infinite recursion or very deep recursion.
- **Memory Leaks:** Dynamically allocated memory that is no longer referenced by the program but has not been deallocated using free().
- **Segmentation Faults:** Occur when a program tries to access a memory location that it is not allowed to access (e.g., dereferencing a NULL pointer, accessing out-of-bounds array memory).
- Returning pointers to local stack variables (dangling pointers).

- **Standard Problem-Solving Techniques or Shortcuts:**

- Draw the stack and heap to visualize memory allocation and deallocation.
- For recursive functions, trace the stack frames to understand memory usage.
- Always match every malloc() with a free() to prevent memory leaks.

Quick Formula Reference

This section consolidates key formulas and results for quick revision.

- **Array Address Calculation (1D):**

$$\text{Address}(A[i]) = \text{BaseAddress} + i \times \text{sizeof}(\text{ElementType})$$

- **Array Address Calculation (2D, Row-Major):**

$$\text{Address}(A[i][j]) = \text{BaseAddress} + (i \times \text{cols} + j) \times \text{sizeof}(\text{ElementType})$$

- **Pointer Arithmetic (Addition):**

$$\text{Address}(p + n) = \text{Address}(p) + n \times \text{sizeof}(*p)$$

- **Pointer Arithmetic (Subtraction):**

$$p - q = \frac{\text{Address}(p) - \text{Address}(q)}{\text{sizeof}(*p)} \quad (\text{for same type pointers})$$

- **sizeof(struct):**

$$\text{sizeof}(\text{struct}) \geq \sum \text{sizeof}(\text{member}) \quad (\text{due to padding})$$

- **sizeof(union):**

$$\text{sizeof}(\text{union}) = \max(\text{sizeof}(\text{member}_1), \text{sizeof}(\text{member}_2), \dots)$$

- **String Length:** `strlen(s)` returns number of characters before `\0`.
- **Integer Division:** a/b truncates towards zero for positive integers. E.g., $5/2 = 2$.
- **Modulo Operator:** $a\%b$ result sign is implementation-defined for negative operands, but typically matches the sign of a .

Important Tips for GATE

1. **Master Pointers and Arrays:** A significant portion of C questions revolves around pointers, array indexing, and their interaction. Practice drawing memory diagrams for complex pointer expressions and array manipulations. Understand array decay to pointers.
2. **Trace Code Meticulously:** For output prediction questions, mentally (or on scratch paper) trace the execution flow, variable values, and memory changes step-by-step. Pay close attention to loop conditions, conditional statements, and function calls.
3. **Understand Scope and Lifetime:** Differentiate between local, global, and static variables. Be aware of when variables are created and destroyed, especially for recursive functions and functions returning pointers to local variables.
4. **Beware of Type Conversions and Operator Precedence:** C's implicit type conversions can lead to unexpected results. Always be mindful of data types in expressions. Memorize common operator precedence rules (e.g., `*` and `/` before `+` and `-`; unary operators have high precedence).
5. **Practice Recursion Tracing:** Recursion questions are common. Practice tracing recursive calls, identifying the base case, and understanding how values are returned up the call stack. Watch out for stack overflow scenarios.
6. **Look for Edge Cases and Undefined Behavior:** GATE questions often test understanding of edge cases (e.g., empty strings, NULL pointers, array bounds) and undefined behavior (e.g., dereferencing NULL, modifying string literals, out-of-bounds access).
7. **Memory Management:** Understand the difference between stack and heap memory. Know how `malloc()`, `calloc()`, `realloc()`, and `free()` work, and the consequences of memory leaks or double-freeing.
8. **Time Management:** C programming questions can be time-consuming due to detailed tracing. If a question seems too complex, try to quickly identify the core concept being tested or make an educated guess based on common pitfalls. Don't get stuck on one question.

5.1

Aliasing (1)

5.1.1 Aliasing: GATE CSE 2000 | Question: 1.16



Aliasing in the context of programming languages refers to

- multiple variables having the same memory location
- multiple variables having the same value
- multiple variables having the same identifier
- multiple uses of the same variable

gategse-2000 programming easy aliasing

Answer key

5.2

Array (13)

5.2.1 Array: GATE CSE 2002 | Question: 2.8



Consider the following declaration of a two-dimensional array in C:

```
char a[100][100];
```

Assuming that the main memory is byte-addressable and that the array is stored starting from memory address 0, the address of `a[40][50]` is:

- 4040
- 4050
- 5040
- 5050

Answer key

5.2.2 Array: GATE CSE 2011 | Question: 22



What does the following fragment of C program print?

```
char c[] = "GATE2011";
char *p = c;
printf("%s", p + p[3] - p[1]);
```

- A. GATE2011 B. E2011 C. 2011 D. 011

Answer key

5.2.3 Array: GATE CSE 2015 | Set 1 | Question: 35



What is the output of the following C code? Assume that the address of x is 2000 (in decimal) and an integer requires four bytes of memory.

```
int main () {
    unsigned int x [4] [3] =
        {{1, 2, 3}, {4, 5, 6}, {7, 8, 9}, {10, 11, 12}};
    printf ("%u, %u, %u", x + 3, *(x + 3), *(x + 2) + 3);
}
```

- A. 2036,2036,2036 B. 2012,4,2204
C. 2036,10,10 D. 2012,4,6

Answer key

5.2.4 Array: GATE CSE 2015 | Set 3 | Question: 30



Consider the following two C code segments. Y and X are one and two dimensional arrays of size n and $n \times n$ respectively, where $2 \leq n \leq 10$. Assume that in both code segments, elements of Y are initialized to 0 and each element $X[i][j]$ of array X is initialized to $i + j$. Further assume that when stored in main memory all elements of X are in same main memory page frame.

Code segment 1 :

```
// initialize elements of Y to 0
// initialize elements of X[i][j] of X to i+j
for (i=0; i<n; i++)
    Y[i] += X[0][i];
```

Code segment 2 :

```
// initialize elements of Y to 0
// initialize elements of X[i][j] of X to i+j
for (i=0; i<n; i++)
    Y[i] += X[i][0];
```

Which of the following statements is/are correct?

- S1: Final contents of array Y will be same in both code segments
S2: Elements of array X accessed inside the for loop shown in code segment 1 are contiguous in main memory
S3: Elements of array X accessed inside the for loop shown in code segment 2 are contiguous in main memory

- A. Only S2 is correct B. Only S3 is correct
C. Only S1 and S2 are correct D. Only S1 and S3 are correct

Answer key

5.2.5 Array: GATE CSE 2015 | Set 3 | Question: 7



Consider the following C program segment.

```
# include <stdio.h>
```

```
int main()
{
    char s1[7] = "1234", *p;
    p = s1 + 2;
    *p = '0';
    printf("%s", s1);
}
```

What will be printed by the program?

- A. 12 B. 120400 C. 1204 D. 1034

gatecse-2015-set3 programming programming-in-c normal array

Answer key

5.2.6 Array: GATE CSE 2019 | Question: 24



Consider the following C program:

```
#include <stdio.h>
int main() {
    int arr[]={1, 2, 3, 4, 5, 6, 7, 8, 9, 0, 1, 2, 5}, *ip=arr+4;
    printf("%d\n", ip[1]);
    return 0;
}
```

The number that will be displayed on execution of the program is _____

gatecse-2019 numerical-answers programming-in-c programming array easy one-mark

Answer key

5.2.7 Array: GATE CSE 2020 | Question: 22



Consider the following C program.

```
#include <stdio.h>
int main () {
    int a[4][5] = {{1, 2, 3, 4, 5},
                  {6, 7, 8, 9, 10},
                  {11, 12, 13, 14, 15},
                  {16, 17, 18, 19, 20}};
    printf("%d\n", *(a+**a+2)+3));
    return(0);
}
```

The output of the program is _____.

gatecse-2020 numerical-answers programming-in-c array one-mark

Answer key

5.2.8 Array: GATE CSE 2021 | Set 2 | Question: 10



Consider the following ANSI C program.

```
#include <stdio.h>
int main()
{
    int arr[4][5];
    int i, j;
    for (i=0; i<4; i++)
    {
        for (j=0; j<5; j++)
        {
            arr[i][j] = 10 * i + j;
        }
    }
    printf("%d", *(arr[1]+9));
    return 0;
}
```

What is the output of the above program?

- A. 14 B. 20 C. 24 D. 30

Answer key

5.2.9 Array: GATE CSE 2022 | Question: 33



What is printed by the following ANSI C program?

```
#include<stdio.h>

int main (int argc, char *argv[])
{
    int a[3][3][3] =
        {{1, 2, 3, 4, 5, 6, 7, 8, 9},
         {10, 11, 12, 13, 14, 15, 16, 17, 18},
         {19, 20, 21, 22, 23, 24, 25, 26, 27}};

    int i = 0, j = 0, k = 0;
    for ( i = 0; i < 3; i ++ ) {
        for ( k = 0; k < 3; k ++ )
            printf("%d", a[i][j][k]);
        printf("\n");
    }
    return 0;
}
```

- 1 2 3
- A. 10 11 12
19 20 21
1 4 7
- B. 10 13 16
19 22 25
1 2 3
- C. 4 5 6
7 8 9
1 2 3
- D. 13 14 15
25 26 27

Answer key

5.2.10 Array: GATE IT 2004 | Question: 58



Consider the following C program which is supposed to compute the transpose of a given 4×4 matrix M . Note that, there is an X in the program which indicates some missing statements. Choose the correct option to replace X in the program.

```
#include<stdio.h>
#define ROW 4
#define COL 4
int M[ROW][COL] = {1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12, 13, 14, 15, 16};
main()
{
    int i, j, t;
    for (i = 0; i < 4; ++i)
    {
```

```

X
}
for (i = 0; i < 4; ++i)
    for (j = 0; j < 4; ++j)
        printf ("%d", M[i][j]);
}

```

A.

```
for(j = 0; j < 4; ++j){
    t = M[i][j];
    M[i][j] = M[j][i];
    M[j][i] = t;
}
```

B.

```
for(j = 0; j < 4; ++j){
    M[i][j] = t;
    t = M[j][i];
    M[j][i] = M[i][j];
}
```

C.

```
for(j = i; j < 4; ++j){
    t = M[i][j];
    M[i][j] = M[j][i];
    M[j][i] = t;
}
```

D.

```
for(j = i; j < 4; ++j){
    M[i][j] = t;
    t = M[j][i];
    M[j][i] = M[i][j];
}
```

gateit-2004 programming easy programming-in-c array

Answer key

5.2.11 Array: GATE IT 2008 | Question: 49



What is the output printed by the following C code?

```

#include <stdio.h>
int main ()
{
    char a [6] = "world";
    int i, j;
    for (i = 0, j = 5; i < j; a [i++] = a [j--]);
    printf ("%s\n", a);
}

```

A. dlrow

B. Null string

C. dlrlld

D. worow

gateit-2008 programming programming-in-c normal array

Answer key

5.2.12 Array: GATE IT 2008 | Question: 51



Consider the C program given below. What does it print?

```

#include <stdio.h>
int main ()
{
    int i, j;
    int a [8] = {1, 2, 3, 4, 5, 6, 7, 8};
    for(i = 0; i < 3; i++) {
        a[i] = a[i] + 1;
        i++;
    }
    i--;
    for (j = 7; j > 4; j--) {
        int i = j/2;
        a[i] = a[i] - 1;
    }
    printf ("%d, %d", i, a[i]);
}

```

A. 2,3

B. 2,4

C. 3,2

D. 3,3

gateit-2008 programming programming-in-c normal array

Answer key

5.2.13 Array: GATE IT 2008 | Question: 52



C program is given below:

```

#include <stdio.h>
int main ()
{

```

```

int i, j;
char a [2] [3] = {{ 'a', 'b', 'c' }, { 'd', 'e', 'f' }};
char b [3] [2];
char *p = *b;
for (i = 0; i < 2; i++) {
    for (j = 0; j < 3; j++) {
        *(p + 2*j + i) = a [i] [j];
    }
}

```

What should be the contents of the array b at the end of the program?

- A. a b
c d
e f
- B. a d
b e
c f
- C. a c
e b
d f
- D. a e
d c
b f

gateit-2008 programming programming-in-c normal array

Answer key

5.3 Functions (2)

5.3.1 Functions: GATE CSE 2015 | Set 3 | Question: 54



Consider the following C program:

```

#include<stdio.h>
int f1(void);
int f2(void);
int f3(void);
int x=10;
int main()
{
    int x=1;
    x += f1() + f2 () + f3() + f2();
    printf("%d", x);
    return 0;
}
int f1() { int x = 25; x++; return x;}
int f2() { static int x = 50; x++; return x;}
int f3() { x *= 10; return x;}

```

The output of the program is _____.

gatecse-2015-set3 programming programming-in-c functions normal numerical-answers

Answer key

5.3.2 Functions: GATE CSE 2024 | Set 2 | Question: 3



Consider the following C program. Assume parameters to a function are evaluated from *right to left*.

```

#include <stdio.h>

int g(int p) { printf("%d", p); return p; }
int h(int q) { printf("%d", q); return q; }

void f(int x, int y) {

```

```

    g(x);
    h(y);
}

int main() {
    f(g(10), h(20));
}

```

Which one of the following options is the CORRECT output of the above C program?

- A. 20101020 B. 10202010 C. 20102010 D. 10201020

gatecse-2024-set2 programming programming-in-c functions one-mark

Answer key

5.4 Goto (2)

5.4.1 Goto: GATE CSE 1989 | Question: 3-i



An unrestricted use of the "go to" statement is harmful because of which of the following reason (s):

- A. It makes it more difficult to verify programs.
 B. It makes programs more inefficient.
 C. It makes it more difficult to modify existing programs.
 D. It results in the compiler generating longer machine code.

gate1989 normal programming goto

Answer key

5.4.2 Goto: GATE CSE 1994 | Question: 1.5



An unrestricted use of the "goto" statement is harmful because

- A. it makes it more difficult to verify programs
 B. it increases the running time of the programs
 C. it increases the memory required for the programs
 D. it results in the compiler generating longer machine code

gate1994 programming easy goto

Answer key

5.5 Identify Function (6)

5.5.1 Identify Function: GATE CSE 1995 | Question: 3



Consider the following high level programming segment. Give the contents of the memory locations for variables W , X , Y and Z after the execution of the program segment. The values of the variables A and B are $5CH$ and $92H$, respectively. Also indicate error conditions if any.

```

var
    A, B, W, X, Y : unsigned byte;
    Z              : unsigned integer, (each integer is represented by two bytes)
begin
    X              := A+B
    Y              := abs(A-B);
    W              := A-B
    Z              := A*B
end;

```

gate1995 programming identify-function descriptive

Answer key

5.5.2 Identify Function: GATE CSE 1998 | Question: 2.13



What is the result of the following program?


```

program side-effect (input, output);
var x, result: integer;
function f (var x:integer):integer;
begin
    x:=x+1;f:=x;
end
begin
    x:=5;
    result:=f(x)*f(x);
    writeln(result);
end

```

A. 5

B. 25

C. 36

D. 42

gate1998 programming normal identify-function

Answer key

5.5.3 Identify Function: GATE CSE 2017 | Set 2 | Question: 14



Consider the following function implemented in C:

```

void printxy(int x, int y) {
    int *ptr;
    x=0;
    ptr=&x;
    y=*ptr;
    *ptr=1;
    printf("%d, %d", x, y);
}

```

The output of invoking *printxy*(1,1) is:

A. 0,0

B. 0,1

C. 1,0

D. 1,1

gatecse-2017-set2 programming-in-c identify-function pointers

Answer key

5.5.4 Identify Function: GATE CSE 2017 | Set 2 | Question: 43



Consider the following snippet of a C program. Assume that swap (&*x*, &*y*) exchanges the content of *x* and *y*:

```

int main () {
    int array[] = {3, 5, 1, 4, 6, 2};
    int done =0;
    int i;
    while (done==0) {
        done =1;
        for (i=0; i<=4; i++) {
            if (array[i] < array[i+1]) {
                swap(&array[i], &array[i+1]);
                done=0;
            }
        }
        for (i=5; i>=1; i--) {
            if (array[i] > array[i-1]) {
                swap(&array[i], &array[i-1]);
                done =0;
            }
        }
    }
    printf("%d", array[3]);
}

```

The output of the program is _____

gatecse-2017-set2 programming algorithms numerical-answers identify-function

Answer key

5.5.5 Identify Function: GATE CSE 2019 | Question: 18



Consider the following C program :

```
#include<stdio.h>
```

```

int jumble(int x, int y){
    x = 2*x+y;
    return x;
}
int main(){
    int x=2, y=5;
    y=jumble(y,x);
    x=jumble(y,x);
    printf("%d \n",x);
    return 0;
}

```

The value printed by the program is _____.

gatecse-2019 programming-in-c numerical-answers identify-function one-mark

Answer key

5.5.6 Identify Function: GATE IT 2004 | Question: 15



Let x be an integer which can take a value of 0 or 1. The statement

```
if (x == 0) x = 1; else x = 0;
```

is equivalent to which one of the following ?

- A. $x = 1 + x$; B. $x = 1 - x$; C. $x = x - 1$; D. $x = 1 \% x$;

gateit-2004 programming easy identify-function

Answer key

5.6 Loop Invariants (8)

5.6.1 Loop Invariants: GATE CSE 1987 | Question: 7a



List the invariant assertions at points A, B, C, D and E in program given below:

```

Program division (input, output)
Const
    dividend = 81;
    divisor = 9;
Var remainder, quotient: interger
begin
    (*dividend >= 0 AND (divisor > 0)*)
    remainder := dividend;
    quotient := 0;
    (*A*)
While (remainder >= 0) do
begin (*B*)
    quotient := quotient + 1;
    remainder := remainder - divisor;
    (*C*)
end;
    (*D*)
    quotient := quotient - 1;
    remainder := remainder + divisor;
    (*E*)
end

```

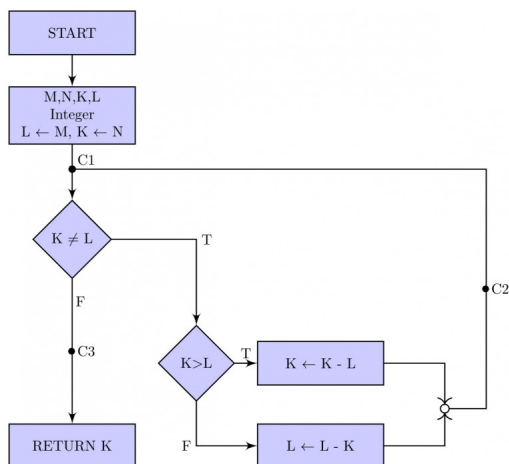
gate1987 programming loop-invariants descriptive

Answer key

5.6.2 Loop Invariants: GATE CSE 1988 | Question: 6ii



Below figure is the flow-chart corresponding to a program to calculate the gcd of two integers, M and N respectively, ($M, N > 0$). Use assertions at the cut point C_1, C_2 and C_3 to prove that the flow-chart is correct.



gate1988 normal descriptive loop-invariants

Answer key

5.6.3 Loop Invariants: GATE CSE 1988 | Question: 8ii



Consider the two program segments below:

a.

```
for
  i:=1 to f(x) by 1 do
  S
end
```

b.

```
i:=1;
While i<=f(x) do
  S
  i:=i+1
end
```

Under what conditions are these two programs equivalent? Treat S as any sequence of statements and f as a function.

gate1988 programming descriptive loop-invariants

Answer key

5.6.4 Loop Invariants: GATE CSE 1991 | Question: 1,vi



Consider the following PASCAL program segment:

```

if i mod 2 = 0 then
  while i >= 0 do
  begin
    i := i div 2;
    if i mod 2 < 0 then i := i - 1;
    else i := i - 2;
  end;

```

An appropriate loop-invariant for the while-loop is _____

gate1991 programming loop-invariants normal fill-in-the-blanks

Answer key

5.6.5 Loop Invariants: GATE CSE 2004 | Question: 32



Consider the following program fragment for reversing the digits in a given integer to obtain a new integer.

Let $n = d_1 d_2 \dots d_m$.

```

int n, rev;
rev = 0;
while(n > 0) {
  rev = rev * 10 + n%10;

```

```

n = n/10;
}

```

The loop invariant condition at the end of the i^{th} iteration is:

- A. $n = d_1 d_2 \dots d_{m-i}$ **and** $rev = d_m d_{m-1} \dots d_{m-i+1}$
 B. $n = d_{m-i+1} \dots d_{m-1} d_m$ **or** $rev = d_{m-i} \dots d_2 d_1$
 C. $n \neq rev$
 D. $n = d_1 d_2 \dots d_m$ **or** $rev = d_m \dots d_2 d_1$

gatecse-2004 programming loop-invariants normal

Answer key

5.6.6 Loop Invariants: GATE CSE 2015 | Set 1 | Question: 33



Consider the following pseudo code, where x and y are positive integers.

```

begin
  q := 0
  r := x
  while r ≥ y do
    begin
      r := r - y
      q := q + 1
    end
  end
end

```

The post condition that needs to be satisfied after the program terminates is

- A. $\{r = qx + y \wedge r < y\}$
 B. $\{x = qy + r \wedge r < y\}$
 C. $\{y = qx + r \wedge 0 < r < y\}$
 D. $\{q + 1 < r - y \wedge y > 0\}$

gatecse-2015-set1 programming loop-invariants normal

Answer key

5.6.7 Loop Invariants: GATE CSE 2016 | Set 2 | Question: 35



The following function computes X^Y for positive integers X and Y .

```

int exp (int X, int Y) {
  int res = 1, a = X, b = Y;

  while (b != 0) {
    if (b % 2 == 0) {a = a * a; b = b/2;}
    else {res = res * a; b = b - 1;}
  }
  return res;
}

```

Which one of the following conditions is TRUE before every iteration of the loop?

- A. $X^Y = a^b$ B. $(res * a)^Y = (res * X)^b$
 C. $X^Y = res * a^b$ D. $X^Y = (res * a)^b$

gatecse-2016-set2 programming loop-invariants normal

Answer key

5.6.8 Loop Invariants: GATE CSE 2017 | Set 2 | Question: 37



Consider the C program fragment below which is meant to divide x by y using repeated subtractions. The variables x , y , q and r are all unsigned int.

```

while (r >= y) {
  r = r - y;
  q = q + 1;
}

```

Which of the following conditions on the variables x, y, q and r before the execution of the fragment will ensure that the loop terminated in a state satisfying the condition $x == (y * q + r)$?

- A. $(q == r) \ \&\& \ (r == 0)$
- B. $(x > 0) \ \&\& \ (r == x) \ \&\& \ (y > 0)$
- C. $(q == 0) \ \&\& \ (r == x) \ \&\& \ (y > 0)$
- D. $(q == 0) \ \&\& \ (y > 0)$

gatecse-2017-set2 programming loop-invariants

Answer key

5.7 Output (8)

5.7.1 Output: GATE CSE 2022 | Question: 34



What is printed by the following ANSI C program?

```
#include<stdio.h>

int main(int argc, char *argv[]) {
    char a = 'P';
    char b = 'x';
    char c = (a&b) + '*';
    char d = (a|b) - '-';
    char e = (a^b) + '+';

    printf("%c %c %c\n", c, d, e);

    return 0;
}
```

ASCII encoding for relevant characters is given below

A	B	C	...	Z
65	66	67	...	90

a	b	c	...	z
97	98	99	...	122

*	+	-
42	43	45

- A. z K S
- B. 122 75 83
- C. * - +
- D. P x +

gatecse-2022 programming programming-in-c output two-marks

Answer key

5.7.2 Output: GATE CSE 2023 | Question: 25



The integer value printed by the ANSI-C program given below is _____

```
#include<stdio.h>

int funcp(){
    static int x = 1;
    x++;
    return x;
}

int main(){
    int x,y;
    x = funcp();
    y = funcp()+x;
    printf("%d\n", (x+y));
    return 0;
}
```

Answer key

5.7.3 Output: GATE CSE 2024 | Set 1 | Question: 8



Consider the following C program:

```
#include <stdio.h>

int main() {
    int a=6;
    int b = 0;
    while (a<10) {
        a = a / 12+1 ;
        a += b ;}
    printf ("%d", a);
    return 0 ; }
```

Which one of the following statements is CORRECT?

- A. The program prints 9 as output
- B. The program prints 10 as output
- C. The program gets stuck in an infinite loop
- D. The program prints 6 as output

Answer key

5.7.4 Output: GATE CSE 2024 | Set 1 | Question: 9



Consider the following C program:

```
#include <stdio.h>
void fX ();
int main(){
    fX();
    return 0 ;}

void fX () {
    char a;
    if ((a=getchar()) != '\n')
        fX();
    if (a != '\n')
        putchar (a); }
```

Assume that the input to the program from the command line is 1234 followed by a newline character. Which one of the following statements is CORRECT?

- A. The program will not terminate
- B. The program will terminate with no output
- C. The program will terminate with 4321 as output
- D. The program will terminate with 1234 as output

Answer key

5.7.5 Output: GATE CSE 2025 | Set 1 | Question: 53



Consider the following C program:

```
#include <stdio.h>
int gate (int n) {
    int d, t, newnum, turn;
```

```

newnum = turn = 0; t=1;
while (n>=t) t *= 10;
t /=10;
while (t>0) {
d = n/t;
n = n%t;
t /= 10;
if (turn) newnum = 10*newnum + d;
turn = (turn + 1) % 2;
}
return newnum;
}
int main () {
printf ("%d", gate(14362));
return 0;
}

```

The value printed by the given C program is _____. (Answer in integer)

gatecse2025-set1 programming-in-c output numerical-answers two-marks

Answer key

5.7.6 Output: GATE CSE 2025 | Set 2 | Question: 23



```

int x=126,y=105;
do {
    if(x>y) x=x-y;
    else y=y-x;
} while(x!=y);
printf("%d",x);

```

The output of the given C code segment is _____. (Answer in integer)

gatecse2025-set2 programming-in-c output numerical-answers one-mark

Answer key

5.7.7 Output: GATE CSE 2025 | Set 2 | Question: 53



Consider the following C program:

```

#include <stdio.h>

int g(int n) {
    return (n+10);
}

int f(int n) {
    return g(n*2);
}

int main() {
    int sum, n;
    sum=0;
    for (n=1; n<3; n++)
        sum += g(f(n));
    printf ("%d", sum);
    return 0;
}

```

The output of the given C program is _____. (Answer in integer)

gatecse2025-set2 programming-in-c output numerical-answers two-marks

Answer key

5.7.8 Output: GATE CSE 2026 | Set 1 | Question: 24



Consider the following program in C:

```

#include <stdio.h>
void func(int i, int j) {
    if(i < j) {
        int i = 0;

```

```

while (i < 10) {
    j += 2;
    i++;
}
printf("%d", i);
}
int main() {
    int i = 9, j = 10;
    func(i, j);
    return 0;
}

```

The output of the program is _____. (answer in integer)

Note: Assume that the program compiles and runs successfully.

gatecse-2026-set1 programming-in-c output numerical-answers one-mark

Answer key

5.8

Parameter Passing (12)

5.8.1 Parameter Passing: GATE CSE 1992 | Question: 10b



Show the activation records and the display structure just after the procedures called at lines marked x and y have started their execution. Be sure to indicate which of the two procedures named A you are referring to.

```

Program Test;
Procedure A;
  Procedure B;
    Procedure A;
      begin
        .....
      end A;
    begin
      y: A;
    end B;
  begin
    B;
  end A;
begin
  x: A;
end Test

```

gate1992 parameter-passing programming runtime-environment normal descriptive

Answer key

5.8.2 Parameter Passing: GATE CSE 1994 | Question: 1.20



In which of the following cases is it possible to obtain different results for call-by-reference and call-by-name parameter passing methods?

- | | |
|--|---|
| A. Passing a constant value as a parameter | B. Passing the address of an array as a parameter |
| C. Passing an array element as a parameter | D. Passing an array |

gate1994 programming parameter-passing easy

Answer key

5.8.3 Parameter Passing: GATE CSE 2001 | Question: 2.17 | UGCNET-AUG2016-III: 21



What is printed by the print statements in the program $P1$ assuming call by reference parameter passing?

```

Program P1()
{
    x = 10;
    y = 3;
    func1(y,x,x);
    print x;
}

```



```

print y;
}

func1(x,y,z)
{
    y = y + 4;
    z = x + y + z
}

```

- A. 10, 3 B. 31, 3 C. 27, 7 D. None of the above

gatecse-2001 programming-in-c parameter-passing normal ugcnetcse-aug2016-paper3

Answer key

5.8.4 Parameter Passing: GATE CSE 2003 | Question: 73



The following program fragment is written in a programming language that allows global variables and does not allow nested declarations of functions.

```

global int i=100, j=5;
void P(x) {
    int i=10;
    print(x+10);
    i=200;
    j=20;
    print (x);
}
main() {P(i+j);}

```

If the programming language uses static scoping and call by need parameter passing mechanism, the values printed by the above program are:

- A. 115, 220 B. 25, 220 C. 25, 15 D. 115, 105

gatecse-2003 compiler-design normal runtime-environment parameter-passing

Answer key

5.8.5 Parameter Passing: GATE CSE 2008 | Question: 60



What is printed by the following C program?

```

int f(int x, int *py, int **ppz)
{
    int y, z;
    **ppz += 1; z = **ppz; // corrected z = *ppz; to z = **ppz;
    *py += 2; y = *py;
    x += 3;
    return x+y+z;
}

void main()
{
    int c, *b, **a;
    c = 4; b = &c; a = &b;
    printf("%d", f(c, b, a));
}

```

- A. 18 B. 19 C. 21 D. 22

gatecse-2008 programming programming-in-c normal parameter-passing

Answer key

5.8.6 Parameter Passing: GATE CSE 2010 | Question: 11



What does the following program print?

```

#include<stdio.h>

void f(int *p, int *q) {
    p=q;
    *p=2;
}

```

```
int i=0, j=1;

int main() {
    f(&i, &j);
    printf("%d %d\n", i, j);
    return 0;
}
```

A. 2 2

B. 2 1

C. 0 1

D. 0 2

gatecse-2010 programming programming-in-c easy parameter-passing

Answer key

5.8.7 Parameter Passing: GATE CSE 2013 | Question: 42



What is the return value of $f(p, p)$, if the value of p is initialized to 5 before the call? Note that the first parameter is passed by reference, whereas the second parameter is passed by value.

```
int f (int &x, int c) {
    c = c - 1;
    if (c==0) return 1;
    x = x + 1;
    return f(x,c) * x;
}
```

gatecse-2013 compiler-design normal marks-to-all numerical-answers parameter-passing runtime-environment

Answer key

5.8.8 Parameter Passing: GATE CSE 2016 | Set 1 | Question: 15



Consider the following C program.

```
#include <stdio.h>
void mystery (int *ptrA, int *ptrB) {
    int *temp;
    temp = ptrB;
    ptrB = ptrA;
    ptrA = temp;
}
int main () {
    int a = 2016, b=0, c= 4, d = 42;
    mystery (&a, &b);
    if (a < c)
        mystery (&c, &a);
    mystery (&a, &d);
    printf("%d\n", a);
}
```

The output of the program is _____.

gatecse-2016-set1 programming-in-c easy numerical-answers parameter-passing

Answer key

5.8.9 Parameter Passing: GATE CSE 2016 | Set 2 | Question: 12



The value printed by the following program is _____.

```
void f (int *p, int m) {
    m = m + 5;
    *p = *p + m;
    return;
}
void main () {
    int i=5, j=10;

    f (&i, j);
    printf ("%d", i+j);
}
```

gatecse-2016-set2 programming-in-c normal numerical-answers parameter-passing

Answer key

5.8.10 Parameter Passing: GATE CSE 2018 | Question: 29



```
#include<stdio.h>
void fun1(char* s1, char* s2){
    char* temp;
    temp = s1;
    s1 = s2;
    s2 = temp;
}
void fun2(char** s1, char** s2){
    char* temp;
    temp = *s1;
    *s1 = *s2;
    *s2 = temp;
}
int main(){
    char *str1="Hi", *str2 = "Bye";
    fun1(str1, str2); printf("%s %s", str1, str2);
    fun2(&str1, &str2); printf("%s %s", str1, str2);
    return 0;
}
```

The output of the program above is:

- A. Hi Bye Bye Hi
- B. Hi Bye Hi Bye
- C. Bye Hi Hi Bye
- D. Bye Hi Bye Hi

gatecse-2018 programming-in-c pointers parameter-passing normal programming two-marks

Answer key

5.8.11 Parameter Passing: GATE IT 2006 | Question: 50



Which one of the choices given below would be printed when the following program is executed?

```
#include <stdio.h>
void swap (int *x, int *y)
{
    static int *temp;
    temp = x;
    x = y;
    y = temp;
}
void printab ()
{
    static int i, a = -3, b = -6;
    i = 0;
    while (i <= 4)
    {
        if ((i++)%2 == 1) continue;
        a = a + i;
        b = b + i;
    }
    swap (&a, &b);
    printf("a = %d, b = %d\n", a, b);
}
main()
{
    printab();
    printab();
}
```

- A. $a = 0, b = 3$
 $a = 0, b = 3$
- B. $a = 3, b = 0$
 $a = 12, b = 9$
- C. $a = 3, b = 6$
 $a = 3, b = 6$
- D. $a = 6, b = 3$
 $a = 15, b = 12$

gateit-2006 programming programming-in-c normal parameter-passing

Answer key

5.8.12 Parameter Passing: GATE IT 2008 | Question: 50



Consider the C program below. What does it print?

```
# include <stdio.h>
```

```
# define swap1 (a, b) tmp = a; a = b; b = tmp
void swap2 ( int a, int b)
{
    int tmp;
    tmp = a; a = b; b = tmp;
}
void swap3 (int*a, int*b)
{
    int tmp;
    tmp = *a; *a = *b; *b = tmp;
}
int main ()
{
    int num1 = 5, num2 = 4, tmp;
    if (num1 < num2) {swap1 (num1, num2);}
    if (num1 < num2) {swap2 (num1 + 1, num2);}
    if (num1 >= num2) {swap3 (&num1, &num2);}
    printf ("%d, %d", num1, num2);
}
```

A. 5,5

B. 5,4

C. 4,5

D. 4,4

gateit-2008 programming programming-in-c easy parameter-passing

Answer key

5.9

Pointers (15)

5.9.1 Pointers: GATE CSE 2000 | Question: 1.12



The most appropriate matching for the following pairs

X : m = malloc(5); m = NULL;	1 : using dangling pointers
Y : free(n); n->value = 5;	2 : using uninitialized pointers
Z : char *p, *p = 'a';	3 : lost memory

is:

A. X – 1 Y – 3 Z – 2

B. X – 2 Y – 1 Z – 3

C. X – 3 Y – 2 Z – 1

D. X – 3 Y – 1 Z – 2

gatecse-2000 programming programming-in-c easy match-the-following pointers

Answer key

5.9.2 Pointers: GATE CSE 2001 | Question: 2.18



Consider the following three C functions:

[P1]

```
int *g(void)
{
    int x = 10;
    return (&x);
}
```

[P2]

```
int *g(void)
{
    int *px;
    *px = 10;
    return px;
}
```

[P3]

```
int *g(void)
{
    int *px;
    px = (int*) malloc (sizeof(int));
    *px = 10;
    return px;
}
```

Which of the above three functions are likely to cause problems with pointers?

- A. Only $P3$ B. Only $P1$ and $P3$ C. Only $P1$ and $P2$ D. $P1, P2$ and $P3$

gatecse-2001 programming programming-in-c normal pointers

Answer key

5.9.3 Pointers: GATE CSE 2003 | Question: 2



Assume the following C variable declaration:

```
int *A[10], B[10][10];
```

Of the following expressions:

- I. $A[2]$
- II. $A[2][3]$
- III. $B[1]$
- IV. $B[2][3]$

which will not give compile-time errors if used as left hand sides of assignment statements in a C program?

- A. I, II, and IV only B. II, III, and IV only C. II and IV only D. IV only

gatecse-2003 programming programming-in-c easy pointers

Answer key

5.9.4 Pointers: GATE CSE 2003 | Question: 89



Consider the C program shown below:

```
#include<stdio.h>
#define print(x) printf("%d", x)

int x;
void Q(int z)
{
    z+=x;
    print(z);
}

void P(int *y)
{
    int x = *y + 2;
    Q(x);
    *y = x - 1;
    print(x);
}

main(void) {
    x = 5;
    P(&x);
    print(x);
}
```

The output of this program is:

- A. 12 7 6 B. 22 12 11 C. 14 6 6 D. 7 6 6

gatecse-2003 programming programming-in-c normal pointers

Answer key

5.9.5 Pointers: GATE CSE 2005 | Question: 1, ISRO2017-55



What does the following C-statement declare?

```
int (*f) (int * );
```

- A. A function that takes an integer pointer as argument and returns an integer
B. A function that takes an integer as argument and returns an integer pointer
C. A pointer to a function that takes an integer pointer as argument and returns an integer

D. A function that takes an integer pointer as argument and returns a function pointer

gatecse-2005 programming programming-in-c pointers easy isro2017

Answer key

5.9.6 Pointers: GATE CSE 2006 | Question: 57



Consider this C code to swap two integers and these five statements: the code

```
void swap (int *px, int *py)
{
    *px = *px - *py;
    *py = *px + *py;
    *px = *py - *px;
}
```

S1: will generate a compilation error

S2: may generate a segmentation fault at runtime depending on the arguments passed

S3: correctly implements the swap procedure for all input pointers referring to integers stored in memory locations accessible to the process

S4: implements the swap procedure correctly for some but not all valid input pointers

S5: may add or subtract integers and pointers

A. S1

B. S2 and S3

C. S2 and S4

D. S2 and S5

gatecse-2006 programming programming-in-c normal pointers

Answer key

5.9.7 Pointers: GATE CSE 2014 | Set 1 | Question: 10



Consider the following program in C language:

```
#include <stdio.h>

main()
{
    int i;
    int* pi = &i;

    scanf("%d", pi);
    printf("%d\n", i+5);
}
```

Which one of the following statements is **TRUE**?

A. Compilation fails.

B. Execution results in a run-time error.

C. On execution, the value printed is 5 more than the address of variable *i*.

D. On execution, the value printed is 5 more than the integer value entered.

gatecse-2014-set1 programming programming-in-c easy pointers

Answer key

5.9.8 Pointers: GATE CSE 2015 | Set 3 | Question: 26



Consider the following C program

```
#include <stdio.h>
int main() {
    static int a[] = {10, 20, 30, 40, 50};
    static int *p[] = {a, a+3, a+4, a+1, a+2};
    int **ptr = p;
    ptr++;
    printf("%d%d", ptr-p, **ptr);
}
```

The output of the program is _____.

Answer key

5.9.9 Pointers: GATE CSE 2017 | Set 1 | Question: 13



Consider the following C code:

```
#include<stdio.h>
int *assignval (int *x, int val) {
    *x = val;
    return x;
}

void main () {
    int *x = malloc(sizeof(int));
    if (NULL == x) return;
    x = assignval (x,0);
    if (x) {
        x = (int *)malloc(sizeof(int));
        if (NULL == x) return;
        x = assignval (x,10);
    }
    printf("%d\n", *x);
    free(x);
}
```

The code suffers from which one of the following problems:

- A. compiler error as the return of *malloc* is not typecast appropriately.
- B. compiler error because the comparison should be made as $x == \text{NULL}$ and not as shown.
- C. compiles successfully but execution may result in dangling pointer.
- D. compiles successfully but execution may result in memory leak.

Answer key

5.9.10 Pointers: GATE CSE 2017 | Set 2 | Question: 55



Consider the following C program.

```
#include<stdio.h>
#include<string.h>
int main() {
    char* c="GATECSIT2017";
    char* p=c;
    printf("%d", (int)strlen(c+2[p]-6[p]-1));
    return 0;
}
```

The output of the program is _____

Answer key

5.9.11 Pointers: GATE CSE 2022 | Question: 11



What is printed by the following ANSI C program?

```
#include<stdio.h>

int main(int argc, char *argv[])
{
    int x = 1, z[2] = {10, 11};
    int *p = NULL;
    p = &x;
    *p = 10;
```

```

p = &z[1];

*(&z[0] + 1) += 3;

printf("%d, %d, %d\n", x, z[0], z[1]);

return 0;
}

```

- A. 1,10,11 B. 1,10,14 C. 10,14,11 D. 10,10,14

gatecse-2022 programming programming-in-c pointers output one-mark

Answer key

5.9.12 Pointers: GATE CSE 2024 | Set 2 | Question: 26



What is the output of the following C program?

```

#include <stdio.h>
int main() {
double a[2]={20.0,25.0}; *p,*q;
p=a ;
q=p+1 ;
printf("%d,%d", (int) (q-p), (int)(*q- *p));
return 0;}

```

- A. 4,8 B. 1,5 C. 8,5 D. 1,8

gatecse-2024-set2 programming programming-in-c pointers two-marks

Answer key

5.9.13 Pointers: GATE CSE 2025 | Set 1 | Question: 24



```

#include <stdio.h>
void foo(int *p, int x) {
*p=x;
}
int main(){
int *z;
int a = 20, b = 25;
z = &a;
foo(z,b);
printf("%d",a);
return 0;
}

```

The output of the given C program is _____. (Answer in integer)

gatecse2025-set1 programming-in-c pointers output numerical-answers easy one-mark

Answer key

5.9.14 Pointers: GATE CSE 2025 | Set 2 | Question: 52



Consider the following C program:

```

#include<stdio.h>
int main(){
int a;
int arr[5] = {30,50,10};
int *ptr;
ptr = &arr[0] + 1;
a = *ptr;
(*ptr)++;
ptr++;
printf("%d", a + (*ptr) + arr[1]);
return 0;
}

```

The output of the above program is _____. (Answer in integer)

Answer key

5.9.15 Pointers: GATE CSE 2026 | Set 2 | Question: 50



Consider the following ANSI-C program.

```
#include <stdio.h>
int main(){
    int *ptr, a, b, c;
    a=5; b=11; c=20;
    ptr=&a; *ptr=c; ptr=&c;
    a=*(&b); c=*ptr-a;
    printf("%d",c);
    return(0);
}
```

The output of this program is _____. (answer in integer)

Note: Assume that the program compiles and runs successfully.

Answer key

5.10

Programming Constructs (1)

5.10.1 Programming Constructs: GATE CSE 1999 | Question: 2.5



Given the programming constructs

- i. assignment
- ii. for loops where the loop parameter cannot be changed within the loop
- iii. if-then-else
- iv. forward go to
- v. arbitrary go to
- vi. non-recursive procedure call
- vii. recursive procedure/function call
- viii. repeat loop,

which constructs will you not include in a programming language such that it should be possible to program the terminates (i.e., halting) function in the same programming language

- | | |
|------------------------|-------------------------|
| A. (ii), (iii), (iv) | B. (v), (vii), (viii) |
| C. (vi), (vii), (viii) | D. (iii), (vii), (viii) |

Answer key

5.11

Programming In C (29)

5.11.1 Programming In C: GATE CSE 2000 | Question: 2.20



The value of j at the end of the execution of the following C program:

```
int incr (int i)
{
    static int count = 0;
    count = count + i;
    return (count);
}
main () {
    int i, j;
    for (i = 0; i <= 4; i++)
        j = incr (i);
}
```

is:

A. 10

B. 4

C. 6

D. 7

gatecse-2000 programming programming-in-c easy

Answer key

5.11.2 Programming In C: GATE CSE 2002 | Question: 1.17



In the C language:

- A. At most one activation record exists between the current activation record and the activation record for the main
- B. The number of activation records between the current activation record and the activation records from the main depends on the actual function calling sequence.
- C. The visibility of global variables depends on the actual function calling sequence
- D. Recursion requires the activation record for the recursive function to be saved in a different stack before the recursive function can be called.

gatecse-2002 programming programming-in-c easy descriptive

Answer key

5.11.3 Programming In C: GATE CSE 2002 | Question: 2.18



The C language is:

- A. A context free language
- B. A context sensitive language
- C. A regular language
- D. Parsable fully only by a Turing machine

gatecse-2002 programming programming-in-c normal

Answer key

5.11.4 Programming In C: GATE CSE 2005 | Question: 32



Consider the following C program:

```
double foo (double); /* Line 1 */
int main() {
    double da, db;
    //input da
    db = foo(da);
}
double foo (double a) {
    return a;
}
```

The above code compiled without any error or warning. If Line 1 is deleted, the above code will show:

- A. no compile warning or error
- B. some compiler-warnings not leading to unintended results
- C. some compiler-warnings due to type-mismatch eventually leading to unintended results
- D. compiler errors

gatecse-2005 programming programming-in-c compiler-design easy

Answer key

5.11.5 Programming In C: GATE CSE 2008 | Question: 18



Which combination of the integer variables x , y , and z makes the variable a get the value 4 in the following expression?

$$a = (x > y) ? ((x > z) ? x : z) : ((y > z) ? y : z)$$

- A. $x = 3, y = 4, z = 2$
- B. $x = 6, y = 5, z = 3$
- C. $x = 6, y = 3, z = 5$
- D. $x = 5, y = 4, z = 5$

gatecse-2008 programming programming-in-c easy

5.11.6 Programming In C: GATE CSE 2008 | Question: 61



Choose the correct option to fill ?1 and ?2 so that the program below prints an input string in reverse order. Assume that the input string is terminated by a new line character.

```
void reverse(void)
{
    int c;
    if(?1) reverse();
    ?2
}
main()
{
    printf("Enter text");
    printf("\n");
    reverse();
    printf("\n");
}
```

- A. ?1 is $(getchar() \neq \backslash n)$
 ?2 is $getchar(c)$;
 B. ?1 is $((c = getchar()) \neq \backslash n)$
 ?2 is $getchar(c)$;
 C. ?1 is $(c \neq \backslash n)$
 ?2 is $putchar(c)$;
 D. ?1 is $((c = getchar()) \neq \backslash n)$
 ?2 is $putchar(c)$;

gatecse-2008 programming normal programming-in-c

5.11.7 Programming In C: GATE CSE 2012 | Question: 48



Consider the following C code segment.

```
int a, b, c = 0;
void prtFun(void);
main()
{
    static int a = 1; /* Line 1 */
    prtFun();
    a += 1;
    prtFun();
    printf("\n %d %d ", a, b);
}

void prtFun(void)
{
    static int a = 2; /* Line 2 */
    int b = 1;
    a += ++b;
    printf("\n %d %d ", a, b);
}
```

What output will be generated by the given code segment?

- | | | | |
|--------|--------|--------|--------|
| A. 3 1 | B. 4 2 | C. 4 2 | D. 3 1 |
| 4 1 | 6 1 | 6 2 | 5 2 |
| 4 2 | 6 1 | 2 0 | 5 2 |

gatecse-2012 programming programming-in-c normal

5.11.8 Programming In C: GATE CSE 2012 | Question: 49



Consider the following C code segment.

```
int a, b, c = 0;
void prtFun(void);
main()
{
    static int a = 1;    /* Line 1 */
    prtFun();
    a += 1;
    prtFun();
    printf("\n %d %d ", a, b);
}

void prtFun(void)
{
    static int a = 2;    /* Line 2 */
    int b = 1;
    a += ++b;
    printf("\n %d %d ", a, b);
}
```

What output will be generated by the given code segment if:

Line 1 is replaced by **auto int a = 1;**

Line 2 is replaced by **register int a = 2;**

- | | | | |
|--------|--------|--------|--------|
| 3 1 | 4 2 | 4 2 | 4 2 |
| A. 4 1 | B. 6 1 | C. 6 2 | D. 4 2 |
| 4 2 | 6 1 | 2 0 | 2 0 |

normal gatecse-2012 programming-in-c programming

Answer key

5.11.9 Programming In C: GATE CSE 2014 | Set 2 | Question: 11



Suppose n and p are unsigned int variables in a C program. We wish to set p to nC_3 . If n is large, which one of the following statements is most likely to set p correctly?

- | | |
|---|---|
| A. $p = n * (n - 1) * (n - 2) / 6;$ | B. $p = n * (n - 1) / 2 * (n - 2) / 3;$ |
| C. $p = n * (n - 1) / 3 * (n - 2) / 2;$ | D. $p = n * (n - 1) * (n - 2) / 6.0;$ |

gatecse-2014-set2 programming programming-in-c normal

Answer key

5.11.10 Programming In C: GATE CSE 2014 | Set 2 | Question: 42



Consider the C function given below.

```
int f(int j)
{
    static int i = 50;
    int k;
    if (i == j)
    {
        printf("something");
        k = f(i);
        return 0;
    }
    else return 0;
}
```

Which one of the following is **TRUE**?

- The function returns 0 for all values of j .
- The function prints the string **something** for all values of j .
- The function returns 0 when $j = 50$.
- The function will exhaust the runtime stack or run into an infinite loop when $j = 50$.

gatecse-2014-set2 programming programming-in-c

Answer key

5.11.11 Programming In C: GATE CSE 2015 | Set 1 | Question: 11



The output of the following C program is_____.

```
void f1 ( int a, int b) {
    int c;
    c = a; a = b;
    b = c;
}
void f2 ( int *a, int *b) {
    int c;
    c = *a; *a = *b; *b = c;
}
int main () {
    int a = 4, b = 5, c = 6;
    f1 ( a, b);
    f2 (&b, &c);
    printf ("%d", c - a - b);
}
```

gatecse-2015-set1 programming programming-in-c easy numerical-answers

Answer key

5.11.12 Programming In C: GATE CSE 2016 | Set 1 | Question: 12



Consider the following "C" program.

```
void f(int, short);
void main()
{
    int i = 100;
    short s = 12;
    short *p = &s;
    _____; // call to f()
}
```

Which one of the following expressions , when placed in the blank above, will NOT result in a type checking error?

- A. $f(s, *s)$ B. $i = f(i, s)$ C. $f(i, *s)$ D. $f(i, *p)$

gatecse-2016-set1 programming-in-c easy

Answer key

5.11.13 Programming In C: GATE CSE 2016 | Set 1 | Question: 34



The following function computes the maximum value contained in an integer array $P[]$ of size n ($n \geq 1$).

```
int max (int *p, int n) {
    int a = 0, b = n-1;

    while (_____) {
        if (p[a] <= p[b]) {a = a+1;}
        else {b = b-1;}
    }
    return p[a];
}
```

The missing loop condition is:

- A. $a \neq n$ B. $b \neq 0$
C. $b > (a + 1)$ D. $b \neq a$

gatecse-2016-set1 programming-in-c normal

Answer key

5.11.14 Programming In C: GATE CSE 2017 | Set 1 | Question: 53



Consider the following C program.

```
#include <stdio.h>
```

```
#include<string.h>

void printlength(char *s, char *t) {
    unsigned int c=0;
    int len = ((strlen(s) - strlen(t)) > c) ? strlen(s) : strlen(t);
    printf("%d\n", len);
}

void main() {
    char *x = "abc";
    char *y = "defgh";
    printlength(x,y);
}
```

Recall that *strlen* is defined in *string.h* as returning a value of type *size_t*, which is an unsigned int. The output of the program is _____ .

gatecse-2017-set1 programming programming-in-c normal numerical-answers

Answer key

5.11.15 Programming In C: GATE CSE 2017 | Set 1 | Question: 55



The output of executing the following C program is _____ .

```
#include<stdio.h>

int total(int v) {
    static int count = 0;
    while(v) {
        count += v&1;
        v >>= 1;
    }
    return count;
}

void main() {
    static int x=0;
    int i=5;
    for(; i>0; i--) {
        x = x + total(i);
    }
    printf("%d\n", x);
}
```

gatecse-2017-set1 programming programming-in-c normal numerical-answers

Answer key

5.11.16 Programming In C: GATE CSE 2017 | Set 2 | Question: 2



Match the following:

<i>P.</i> static char var ;	i. Sequence of memory locations to store addresses
<i>Q.</i> m = malloc(10); m=NULL ;	ii. A variable located in data section of memory
<i>R.</i> char *ptr[10] ;	iii. Request to allocate a CPU register to store data
<i>S.</i> register int varl;	iv. A lost memory which cannot be freed

- A. P-ii; Q-iv; R-i; S-iii
C. P-ii; Q-iv; R-iii; S-i

- B. P-ii; Q-i; R-iv; S-iii
D. P-iii; Q-iv; R-i; S-ii

gatecse-2017-set2 programming programming-in-c match-the-following

Answer key

5.11.17 Programming In C: GATE CSE 2017 | Set 2 | Question: 54



Consider the following C program.

```
#include<stdio.h>

int main () {
    int m=10;
    int n, n1;
    n=++m;
```

```

n1=m++;
n--;
--n1;
n-=n1;
printf("%d", n);
return 0;
}

```

The output of the program is _____

gatecse-2017-set2 programming-in-c numerical-answers easy

Answer key

5.11.18 Programming In C: GATE CSE 2018 | Question: 32



Consider the following C code. Assume that unsigned long int type length is 64 bits.

```

unsigned long int fun(unsigned long int n) {
    unsigned long int i, j=0, sum = 0;
    for( i=n; i>1; i=i/2) j++;
    for( ; j>1; j=j/2) sum++;
    return sum;
}

```

The value returned when we call fun with the input 2^{40} is:

- A. 4 B. 5 C. 6 D. 40

gatecse-2018 programming-in-c normal programming two-marks

Answer key

5.11.19 Programming In C: GATE CSE 2019 | Question: 27



Consider the following C program:

```

#include <stdio.h>
int r() {
    static int num=7;
    return num--;
}
int main() {
    for (r();r();r())
        printf("%d",r());
    return 0;
}

```

Which one of the following values will be displayed on execution of the programs?

- A. 41 B. 52 C. 63 D. 630

gatecse-2019 programming-in-c programming two-marks

Answer key

5.11.20 Programming In C: GATE CSE 2019 | Question: 52



Consider the following C program:

```

#include <stdio.h>
int main() {
    float sum = 0.0, j=1.0, i=2.0;
    while (i/j > 0.0625) {
        j=j+i;
        sum=sum+i/j;
        printf("%f\n", sum);
    }
    return 0;
}

```

The number of times the variable sum will be printed, when the above program is executed, is _____

gatecse-2019 numerical-answers programming-in-c programming two-marks

Answer key

5.11.21 Programming In C: GATE CSE 2019 | Question: 53



Consider the following C program:

```
#include <stdio.h>
int main()
{
    int a[] = {2, 4, 6, 8, 10};
    int i, sum=0, *b=a+4;
    for (i=0; i<5; i++)
        sum=sum+(*b-i)-*(b-i);
    printf("%d\n", sum);
    return 0;
}
```

The output of the above C program is _____

gatecse-2019 numerical-answers programming-in-c programming two-marks

Answer key

5.11.22 Programming In C: GATE CSE 2021 | Set 1 | Question: 37



Consider the following ANSI C program.

```
#include <stdio.h>
int main()
{
    int i, j, count;
    count=0;
    i=0;
    for (j=-3; j<=3; j++)
    {
        if ((j >= 0) && (i++))
            count = count + j;
    }
    count = count + i;
    printf("%d", count);
    return 0;
}
```

Which one of the following options is correct?

- A. The program will not compile successfully
- B. The program will compile successfully and output 10 when executed
- C. The program will compile successfully and output 8 when executed
- D. The program will compile successfully and output 13 when executed

gatecse-2021-set1 programming-in-c two-marks

Answer key

5.11.23 Programming In C: GATE CSE 2024 | Set 1 | Question: 38



Consider the following C function definition.

```
int f (int x, int y){
    for (int i=0 ; i<y ; i++ ) {
        x= x + x + y;
    }
    return x;
}
```

Which of the following statements is/are TRUE about the above function?

- A. If the inputs are $x = 20, y = 10$, then the return value is greater than 2^{20}
- B. If the inputs are $x = 20, y = 20$, then the return value is greater than 2^{20}
- C. If the inputs are $x = 20, y = 10$, then the return value is less than 2^{10}
- D. If the inputs are $x = 10, y = 20$, then the return value is greater than 2^{20}

gatecse-2024-set1 multiple-selects programming programming-in-c two-marks

5.11.24 Programming In C: GATE CSE 2026 | Set 2 | Question: 9



Consider the following three ANSI-C programs, P1, P2, and P3.

P1	P2	P3
<pre>#include <stdio.h> int a=5; int main(){ int a=7; return(0); }</pre>	<pre>#include <stdio.h> int main(){ int a=5; int a=7; return(0); }</pre>	<pre>#include <stdio.h> int main(){ int a=5; float a=7; return(0); }</pre>

Which one of the following statements is true?

- A. Only P1 will compile without any error
- B. Only P2 will compile without any error
- C. Only P3 will compile without any error
- D. All three programs P1, P2, and P3 will compile without any error

gatecse-2026-set2 programming-in-c one-mark

5.11.25 Programming In C: GATE IT 2004 | Question: 59



What is the output of the following program?

```
#include<stdio.h>
int funcf (int x);
int funcg (int y);
main ()
{
    int x = 5, y = 10, count;
    for (count = 1; count <= 2; ++count) {
        y += funcf(x) + funcg(x);
        printf ("%d", y);
    }
}
funcf (int x) {
    int y;
    y = funcg(x);
    return (y);
}
funcg (int x) {
    static int y = 10;
    y += 1;
    return (y + x);
}
```

A. 43 80

B. 42 74

C. 33 37

D. 32 32

gateit-2004 programming programming-in-c normal

5.11.26 Programming In C: GATE IT 2004 | Question: 60



Choose the correct option to fill the ?1 and ?2 so that the program prints an input string in reverse order. Assume that the input string is terminated by a new line character.

```
#include <stdio.h>
void wrt_it (void);
int main (void)
{
    printf("Enter Text");
```

```

printf("\n");
wrt_it();
printf("\n");
return 0;
}
void wrt_it(void)
{
    int c;
    if (?1)
        wrt_it();
    ?2
}

```

- A. ?1 is `getchar() != '\n'`
?2 is `getchar(c);`
- B. ?1 is `(c = getchar()) != '\n'`
?2 is `getchar(c);`
- C. ?1 is `c != '\n'`
?2 is `putchar(c);`
- D. ?1 is `(c = getchar()) != '\n'`
?2 is `putchar(c);`

gateit-2004 programming programming-in-c normal

Answer key

5.11.27 Programming In C: GATE IT 2005 | Question: 58



Let a be an array containing n integers in increasing order. The following algorithm determines whether there are two distinct numbers in the array whose difference is a specified number $S > 0$.

```

i = 0; j = 1;
while (j < n) {
    if (E) j++;
    else if (a[j] - a[i] == S) break;
    else i++;
}
if (j < n) printf("yes") else printf("no");

```

Choose the correct expression for E.

- A. $a[j] - a[i] > S$
- B. $a[j] - a[i] < S$
- C. $a[i] - a[j] < S$
- D. $a[i] - a[j] > S$

gateit-2005 programming normal programming-in-c

Answer key

5.11.28 Programming In C: GATE IT 2006 | Question: 51



Which one of the choices given below would be printed when the following program is executed?

```

#include <stdio.h>
int a1[] = {6, 7, 8, 18, 34, 67};
int a2[] = {23, 56, 28, 29};
int a3[] = {-12, 27, -31};
int *x[] = {a1, a2, a3};
void print(int *a[])
{
    printf("%d,", a[0][2]);
    printf("%d,", *a[2]);
    printf("%d,", ++a[0]);
    printf("%d,", *(++a[0]));
    printf("%d\n", a[-1][+1]);
}
main()
{
    print(x);
}

```

- A. 8, -12, 7, 23, 8
- B. 8, 8, 7, 23, 7
- C. -12, -12, 27, -31, 23
- D. -12, -12, 27, -31, 56

gateit-2006 programming programming-in-c normal

5.11.29 Programming In C: GATE IT 2007 | Question: 31



Consider the C program given below :

```
#include <stdio.h>
int main () {
    int sum = 0, maxsum = 0, i, n = 6;
    int a [] = {2, -2, -1, 3, 4, 2};
    for (i = 0; i < n; i++) {
        if (i == 0 || a[i] < 0 || a[i] < a[i - 1]) {
            if (sum > maxsum) maxsum = sum;
            sum = (a[i] > 0) ? a[i] : 0;
        }
        else sum += a[i];
    }
    if (sum > maxsum) maxsum = sum;
    printf ("%d\n", maxsum);
}
```

What is the value printed out when this program is executed?

- A. 9 B. 8 C. 7 D. 6

gateit-2007 programming programming-in-c normal

5.12

Programming Paradigms (2)

5.12.1 Programming Paradigms: GATE CSE 2004 | Question: 1



The goal of structured programming is to:

- A. have well indented programs
 B. be able to infer the flow of control from the compiled code
 C. be able to infer the flow of control from the program text
 D. avoid the use of GOTO statements

gatecse-2004 programming easy programming-paradigms

5.12.2 Programming Paradigms: GATE CSE 2004 | Question: 90



Choose the best matching between the programming styles in Group 1 and their characteristics in Group 2.

Group 1	Group 2
P. Functional	1. Common-based, procedural
Q. Logic	2. Imperative, abstract data types
R. Object-oriented	3. Side-effect free, declarative, expression evaluations
S. Imperative	4. Declarative, clausal representation, theorem proving

- A. $P - 2$ $Q - 3$ $R - 4$ $S - 1$ B. $P - 4$ $Q - 3$ $R - 2$ $S - 1$
 C. $P - 3$ $Q - 4$ $R - 1$ $S - 2$ D. $P - 3$ $Q - 4$ $R - 2$ $S - 1$

gatecse-2004 programming normal programming-paradigms match-the-following

5.13

Recursion (19)

5.13.1 Recursion: GATE CSE 1991 | Question: 01,x



Consider the following recursive definition of *fib*:

```
fib(n) := if n = 0 then 1
        else if n = 1 then 1
```

```
else fib(n-1) + fib(n-2)
```

The number of times *fib* is called (including the first call) for evaluation of *fib*(7) is _____.

gate1991 programming recursion normal numerical-answers

Answer key

5.13.2 Recursion: GATE CSE 1994 | Question: 21



Consider the following recursive function:

```
function fib (n:integer);integer;  
begin  
if (n=0) or (n=1) then fib := 1  
else fib := fib(n-1) + fib(n-2)  
end;
```

The above function is run on a computer with a stack of 64 bytes. Assuming that only return address and parameter are passed on the stack, and that an integer value and an address takes 2 bytes each, estimate the maximum value of *n* for which the stack will not overflow. Give reasons for your answer.

gate1994 programming recursion normal descriptive

Answer key

5.13.3 Recursion: GATE CSE 2000 | Question: 16



A recursive program to compute Fibonacci numbers is shown below. Assume you are also given an array *f*[0 . . . *m*] with all elements initialized to 0.

```
fib(n) {  
  if (n > M) error ();  
  if (n == 0) return 1;  
  if (n == 1) return 1;  
  if (□) (1)  
    return □ (2)  
  t = fib(n - 1) + fib(n - 2);  
  □ (3)  
  return t;  
}
```

- Fill in the boxes with expressions/statement to make *fib*() store and reuse computed Fibonacci values. Write the box number and the corresponding contents in your answer book.
- What is the time complexity of the resulting program when computing *fib*(*n*)?

gatecse-2000 algorithms normal descriptive recursion

Answer key

5.13.4 Recursion: GATE CSE 2001 | Question: 13



Consider the following C program:

```
void abc(char*s)  
{  
  if(s[0]=='\0')return;  
  abc(s+1);  
  abc(s+1);  
  printf("%c",s[0]);  
}  
  
main()  
{  
  abc("123");  
}
```

- What will be the output of the program?
- If *abc*(*s*) is called with a null-terminated string *s* of length *n* characters (not counting the null ('\0') character), how many characters will be printed by *abc*(*s*)?

Answer key

5.13.5 Recursion: GATE CSE 2002 | Question: 11



The following recursive function in C is a solution to the Towers of Hanoi problem.

```
void move(int n, char A, char B, char C) {
    if (.....) {
        move (.....);
        printf("Move disk %d from pole %c to pole %c\n", n, A, C);
        move (.....);
    }
}
```

Fill in the dotted parts of the solution.

Answer key

5.13.6 Recursion: GATE CSE 2004 | Question: 31, ISRO2008-40



Consider the following C function:

```
int f(int n)
{
    static int i = 1;
    if(n >= 5) return n;
    n = n+i;
    i++;
    return f(n);
}
```

The value returned by $f(1)$ is:

- A. 5 B. 6 C. 7 D. 8

Answer key

5.13.7 Recursion: GATE CSE 2005 | Question: 81b



```
double foo(int n)
{
    int i;
    double sum;
    if(n == 0)
    {
        return 1.0;
    }
    else
    {
        sum = 0.0;
        for(i = 0; i < n; i++)
        {
            sum += foo(i);
        }
        return sum;
    }
}
```

Suppose we modify the above function $foo()$ and stores the value of $foo(i)$ $0 \leq i < n$, as and when they are computed. With this modification the time complexity for function $foo()$ is significantly reduced. The space complexity of the modified function would be:

- A. $O(1)$ B. $O(n)$ C. $O(n^2)$ D. $n!$

Answer key

5.13.8 Recursion: GATE CSE 2007 | Question: 42



Consider the following C function:

```
int f(int n)
{
    static int r = 0;
    if (n <= 0) return 1;
    if (n > 3)
    {
        r = n;
        return f(n-2) + 2;
    }
    return f(n-1) + r;
}
```

What is the value of $f(5)$?

- A. 5 B. 7 C. 9 D. 18

gatecse-2007 programming recursion normal

Answer key

5.13.9 Recursion: GATE CSE 2014 | Set 2 | Question: 40



Consider the following function.

```
double f(double x){
    if( abs(x*x - 3) < 0.01)
        return x;
    else
        return f(x/2 + 1.5/x);
}
```

Give a value q (to 2 decimals) such that $f(q)$ will return q :_____.

gatecse-2014-set2 programming recursion numerical-answers normal

Answer key

5.13.10 Recursion: GATE CSE 2015 | Set 2 | Question: 15



Consider the following function written in the C programming language :

```
void foo(char *a)
{
    if (*a && *a != ' ')
    {
        foo(a+1);
        putchar(*a);
    }
}
```

The output of the above function on input "ABCD EFGH" is

- A. ABCD EFGH B. ABCD C. HGFE DCBA D. DCBA

gatecse-2015-set2 programming programming-in-c normal recursion

Answer key

5.13.11 Recursion: GATE CSE 2016 | Set 1 | Question: 35



What will be the output of the following C program?

```
void count (int n) {
    static int d=1;

    printf ("%d",n);
    printf ("%d",d);
    d++;
    if (n>1) count (n-1);
    printf ("%d",d);
}
```

```
void main(){
    count (3);
}
```

- A. 3 1 2 2 1 3 4 4 4
- B. 3 1 2 1 1 1 2 2 2
- C. 3 1 2 2 1 3 4
- D. 3 1 2 1 1 1 2

gatecse-2016-set1 programming-in-c recursion normal

Answer key

5.13.12 Recursion: GATE CSE 2016 | Set 2 | Question: 37



Consider the following program:

```
int f (int * p, int n)
{
    if (n <= 1) return 0;
    else return max (f (p+1, n-1), p[0] - p[1]);
}
int main ()
{
    int a[] = {3, 5, 2, 6, 4};
    printf(" %d", f(a, 5));
}
```

Note: $\max(x, y)$ returns the maximum of x and y .

The value printed by this program is _____.

gatecse-2016-set2 programming-in-c normal numerical-answers recursion

Answer key

5.13.13 Recursion: GATE CSE 2017 | Set 1 | Question: 35



Consider the following two functions.

```
void fun1(int n) {
    if(n == 0) return;
    printf("%d", n);
    fun2(n - 2);
    printf("%d", n);
}
void fun2(int n) {
    if(n == 0) return;
    printf("%d", n);
    fun1(++n);
    printf("%d", n);
}
```

The output printed when `fun1(5)` is called is

- A. 53423122233445
- B. 53423120112233
- C. 53423122132435
- D. 53423120213243

gatecse-2017-set1 programming normal tricky recursion

Answer key

5.13.14 Recursion: GATE CSE 2017 | Set 1 | Question: 36



Consider the C functions `foo` and `bar` given below:

```
int foo(int val) {
    int x=0;
    while(val > 0) {
        x = x + foo(val--);
    }
}
```

```

}
return val;
}

```

```

int bar(int val) {
    int x = 0;
    while(val > 0) {
        x = x + bar(val-1);
    }
    return val;
}

```

Invocations of *foo*(3) and *bar*(3) will result in:

- A. Return of 6 and 6 respectively.
- B. Infinite loop and abnormal termination respectively.
- C. Abnormal termination and infinite loop respectively.
- D. Both terminating abnormally.

gatecse-2017-set1 programming-in-c programming normal recursion

Answer key

5.13.15 Recursion: GATE CSE 2018 | Question: 21



Consider the following C program:

```

#include<stdio.h>

int counter=0;

int calc (int a, int b) {
    int c;
    counter++;
    if(b==3) return (a*a*a);
    else {
        c = calc(a, b/3);
        return (c*c*c);
    }
}

int main() {
    calc(4, 81);
    printf("%d", counter);
}

```

The output of this program is _____.

gatecse-2018 programming-in-c numerical-answers recursion programming one-mark

Answer key

5.13.16 Recursion: GATE CSE 2020 | Question: 46



Consider the following C functions.

```

int fun1(int n) {
    static int i= 0;
    if (n > 0) {
        ++i;
        fun1(n-1);
    }
    return (i);
}

```

```

int fun2(int n) {
    static int i= 0;
    if (n>0) {
        i = i+ fun1 (n) ;
        fun2(n-1) ;
    }
    return (i);
}

```

The return value of *fun2*(5) is _____

gatecse-2020 numerical-answers programming-in-c recursion two-marks

Answer key

5.13.17 Recursion: GATE CSE 2025 | Set 1 | Question: 51



```

#include <stdio.h>

```



```

int foo(int S[],int size) {
    if(size == 0) return 0;
    if(size == 1) return 1;
    if(S[0] != S[1]) return 1+foo(S+1,size-1);
    return foo(S+1,size-1);
}
int main() {
    int A[]={0,1,2,2,2,0,0,1,1};
    printf("%d",foo(A,9));
    return 0;
}

```

The value printed by the given C program is _____. (Answer in integer)

gatecse2025-set1 programming-in-c recursion output numerical-answers two-marks

Answer key

5.13.18 Recursion: GATE CSE 2026 | Set 2 | Question: 51



Consider the following ANSI-C function.

```

int func(int start, int end){
    int length=end+1-start;
    if((length<1)||((start<0)||((end<0)))){ return(0); }
    if(length%3==0){
        return(func(start+1, end));
    }else if(length%3==1){
        return(1+func(start, end-1));
    }else {
        return(func(start+2, end));
    }
}

```

The maximum possible value that can be returned from this function is _____. (answer in integer)

Note: Ignore syntax errors (if any) in the function.

gatecse-2026-set2 programming-in-c recursion output numerical-answers two-marks

Answer key

5.13.19 Recursion: GATE IT 2007 | Question: 27



The function f is defined as follows:

```

int f (int n) {
    if (n <= 1) return 1;
    else if (n % 2 == 0) return f(n/2);
    else return f(3n - 1);
}

```

Assuming that arbitrarily large integers can be passed as a parameter to the function, consider the following statements.

- The function f terminates for finitely many different values of $n \geq 1$.
- The function f terminates for infinitely many different values of $n \geq 1$.
- The function f does not terminate for finitely many different values of $n \geq 1$.
- The function f does not terminate for infinitely many different values of $n \geq 1$.

Which one of the following options is true of the above?

- A. i and iii B. i and iv C. ii and iii D. ii and iv

gateit-2007 programming recursion normal

Answer key

5.14 Runtime Environment (1)

5.14.1 Runtime Environment: GATE CSE 2026 | Set 2 | Question: 7



In C runtime environment, which one of the following is stored in heap?

- A. A static variable declared inside a function
- B. An array of integers declared inside a function
- C. A dynamically allocated array of integers created using malloc() function call
- D. Return address of a function

gatecse-2026-set2 programming-in-c runtime-environment one-mark

Answer key

5.15

Strings (2)

5.15.1 Strings: GATE CSE 2004 | Question: 33



Consider the following C program segment:

```
char p[20]; int i;
char* s = "string";
int length = strlen(s);
for(i = 0; i < length; i++)
    p[i] = s[length-i];
printf("%s", p);
```

The output of the program is:

- A. gnirts
- B. string
- C. gnirt
- D. no output is printed

gatecse-2004 programming programming-in-c strings easy

Answer key

5.15.2 Strings: GATE CSE 2025 | Set 2 | Question: 9



Consider the following C program:

```
#include <stdio.h>
void stringcopy (char *, char *);
int main() {
    char a[30] = "@#Hello World!";
    stringcopy(a, a+2);
    printf("%s\n", a);
    return 0;
}
void stringcopy(char *s, char *t) {
    while (*t)
        *s++ = *t++;
}
```

Which one of the following will be the output of the program?

- A. @#Hello World!
- B. Hello World!
- C. ello World!
- D. Hello Worldld!

gatecse2025-set2 programming-in-c strings output one-mark

Answer key

5.16

Structure (5)

5.16.1 Structure: GATE CSE 2000 | Question: 1.11



The following C declarations:

```
struct node {
    int i;
    float j;
};
struct node *s[10];
```

define s to be:

- A. An array, each element of which is a pointer to a structure of type node
- B. A structure of 2 fields, each field being a pointer to an array of 10 elements
- C. A structure of 3 fields: an integer, a float, and an array of 10 elements
- D. An array, each element of which is a structure of type node

Answer key

5.16.2 Structure: GATE CSE 2018 | Question: 2



Consider the following C program:

```
#include<stdio.h>
struct Ournode{
    char x, y, z;
};
int main() {
    struct Ournode p={'1', '0', 'a'+2};
    struct Ournode *q=&p;
    printf("%c, %c", *((char*)q+1), *((char*)q+2));
    return 0;
}
```

The output of this program is:

- A. 0, c B. 0, a+2 C. '0', 'a+2' D. '0', 'c'

Answer key

5.16.3 Structure: GATE CSE 2021 | Set 2 | Question: 35



Consider the following ANSI C program:

```
#include <stdio.h>
#include <stdlib.h>
struct Node{
    int value;
    struct Node *next;
};
int main() {
    struct Node *boxE, *head, *boxN; int index=0;
    boxE=head= (struct Node *) malloc(sizeof(struct Node));
    head → value = index;
    for (index =1; index<=3; index++){
        boxN = (struct Node *) malloc (sizeof(struct Node));
        boxE → next = boxN;
        boxN → value = index;
        boxE = boxN; }
    for (index=0; index<=3; index++) {
        printf("Value at index %d is %d\n", index, head → value);
        head = head → next;
        printf("Value at index %d is %d\n", index+1, head → value); } }
```

Which one of the following statements below is correct about the program?

- A. Upon execution, the program creates a linked-list of five nodes
 B. Upon execution, the program goes into an infinite loop
 C. It has a missing **return** which will be reported as an error by the compiler
 D. It dereferences an uninitialized pointer that may result in a run-time error

Answer key

5.16.4 Structure: GATE IT 2004 | Question: 61



Consider the following C program:

```
#include <stdio.h>
typedef struct {
    char *a;
    char *b;
} t;
void f1 (t s);
void f2 (t *p);
main()
```

```

{
    static t s = {"A", "B"};
    printf ("%s %s\n", s.a, s.b);
    f1(s);
    printf ("%s %s\n", s.a, s.b);
    f2(&s);
}
void f1 (t s)
{
    s.a = "U";
    s.b = "V";
    printf ("%s %s\n", s.a, s.b);
    return;
}
void f2(t *p)
{
    p -> a = "V";
    p -> b = "W";
    printf ("%s %s\n", p -> a, p -> b);
    return;
}

```

What is the output generated by the program ?

- | | |
|---|---|
| <p>A. <i>A B</i>
<i>U V</i>
<i>V W</i>
<i>V W</i></p> | <p>B. <i>A B</i>
<i>U V</i>
<i>A B</i>
<i>V W</i></p> |
| <p>C. <i>A B</i>
<i>U V</i>
<i>U V</i>
<i>V W</i></p> | <p>D. <i>A B</i>
<i>U V</i>
<i>V W</i>
<i>U V</i></p> |

gateit-2004 programming programming-in-c normal structure

Answer key

5.16.5 Structure: GATE IT 2006 | Question: 49



Which one of the choices given below would be printed when the following program is executed ?

```

#include <stdio.h>
struct test {
    int i;
    char *c;
}st[] = {5, "become", 4, "better", 6, "jungle", 8, "ancestor", 7, "brother"};
main ()
{
    struct test *p = st;
    p += 1;
    ++p -> c;
    printf ("%s", p++ -> c);
    printf ("%c", *++p -> c);
    printf ("%d", p[0].i);
    printf ("%s \n", p -> c);
}

```

- | | |
|----------------------------------|--------------------------------|
| <p>A. jungle, n, 8, nclastor</p> | <p>B. etter, u, 6, ungle</p> |
| <p>C. cetter, k, 6, jungle</p> | <p>D. etter, u, 8, ncestor</p> |

gateit-2006 programming programming-in-c normal structure

Answer key

5.17

Switch Case (2)

5.17.1 Switch Case: GATE CSE 2012 | Question: 3



What will be the output of the following C program segment?

```

char inChar = 'A';
switch ( inChar ) {
    case 'A' : printf ("Choice A \n");
    case 'B' :
    case 'C' : printf ("Choice B");
    case 'D' :
    case 'E' :
    default : printf ("No Choice");
}

```

}
}

- A. No Choice
C. Choice A
Choice B No Choice
- B. Choice A
D. Program gives no output as it is erroneous

gatecse-2012 programming easy programming-in-c switch-case marks-to-all

Answer key

5.17.2 Switch Case: GATE CSE 2015 | Set 3 | Question: 48



Consider the following C program:

```
#include<stdio.h>
int main()
{
    int i, j, k = 0;
    j=2 * 3 / 4 + 2.0 / 5 + 8 / 5;
    k=-j;
    for (i=0; i<5; i++)
    {
        switch(i+k)
        {
            case 1:
            case 2: printf("\n%d", i+k);
            case 3: printf("\n%d", i+k);
            default: printf("\n%d", i+k);
        }
    }
    return 0;
}
```

The number of times printf statement is executed is _____.

gatecse-2015-set3 programming programming-in-c switch-case normal numerical-answers

Answer key

5.18

Type Checking (1)

5.18.1 Type Checking: GATE CSE 2003 | Question: 24



Which of the following statements is FALSE?

- A. In statically typed languages, each variable in a program has a fixed type
B. In un-typed languages, values do not have any types
C. In dynamically typed languages, variables have no types
D. In all statically typed languages, each variable in a program is associated with values of only a single type during the execution of the program

gatecse-2003 programming normal type-checking

Answer key

5.19

Union (1)

5.19.1 Union: GATE CSE 2000 | Question: 1.17, ISRO2015-79



Consider the following C declaration:

```
struct {
    short x[5];
    union {
        float y;
        long z;
    } u;
} t;
```

Assume that the objects of the type short, float and long occupy 2 bytes, 4 bytes and 8 bytes, respectively. The memory requirement for variable t , ignoring alignment consideration, is:

- A. 22 bytes
B. 14 bytes
C. 18 bytes
D. 10 bytes

Answer key

5.20

Variable Binding (1)

5.20.1 Variable Binding: GATE IT 2007 | Question: 34, UGCNET-Dec2012-III: 52



Consider the program below in a hypothetical programming language which allows global variables and a choice of static or dynamic scoping.

```
int i;
program main()
{
    i = 10;
    call f();
}

procedure f()
{
    int i = 20;
    call g ();
}

procedure g()
{
    print i;
}
```

Let x be the value printed under static scoping and y be the value printed under dynamic scoping. Then, x and y are:

- A. $x = 10, y = 20$ B. $x = 20, y = 10$ C. $x = 10, y = 10$ D. $x = 20, y = 20$

Answer key

Answer Keys

5.1.1	A	5.2.1	B	5.2.2	C	5.2.3	A	5.2.4	C
5.2.5	C	5.2.6	6	5.2.7	19	5.2.8	C	5.2.9	A
5.2.10	C	5.2.11	B	5.2.12	C	5.2.13	B	5.3.1	230
5.3.2	A	5.4.1	A	5.4.2	A	5.5.1	N/A	5.5.2	C
5.5.3	C	5.5.4	3	5.5.5	26	5.5.6	B	5.6.1	N/A
5.6.2	N/A	5.6.3	N/A	5.6.4	N/A	5.6.5	A	5.6.6	B
5.6.7	C	5.6.8	C	5.7.1	A	5.7.2	7	5.7.3	C
5.7.4	C	5.7.5	46:46	5.7.6	21:21	5.7.7	46:46	5.7.8	09 :09
5.8.1	N/A	5.8.2	C	5.8.3	B	5.8.4	D	5.8.5	B
5.8.6	D	5.8.7	6561	5.8.8	2016	5.8.9	30	5.8.10	A
5.8.11	D	5.8.12	C	5.9.1	D	5.9.2	C	5.9.3	A
5.9.4	A	5.9.5	C	5.9.6	C	5.9.7	D	5.9.8	140
5.9.9	D	5.9.10	2	5.9.11	D	5.9.12	B	5.9.13	25:25
5.9.14	111:111	5.9.15	9:9	5.10.1	B	5.11.1	A	5.11.2	B
5.11.3	B	5.11.4	D	5.11.5	A	5.11.6	D	5.11.7	C
5.11.8	D	5.11.9	B	5.11.10	D	5.11.11	-5	5.11.12	D
5.11.13	D	5.11.14	3	5.11.15	23	5.11.16	A	5.11.17	0
5.11.18	B	5.11.19	B	5.11.20	5	5.11.21	10	5.11.22	B

5.11.23	B;D
5.11.28	A
5.13.2	16
5.13.7	B
5.13.12	3
5.13.17	5:5
5.15.2	D
5.16.5	B
5.20.1	A

5.11.24	A
5.11.29	C
5.13.3	N/A
5.13.8	D
5.13.13	A
5.13.18	1:1
5.16.1	A
5.17.1	X

5.11.25	A
5.12.1	C
5.13.4	N/A
5.13.9	1.72 : 1.74
5.13.14	C
5.13.19	D
5.16.2	A
5.17.2	10

5.11.26	D
5.12.2	D
5.13.5	N/A
5.13.10	D
5.13.15	4
5.14.1	C
5.16.3	D
5.18.1	C

5.11.27	B
5.13.1	41
5.13.6	C
5.13.11	A
5.13.16	55
5.15.1	D
5.16.4	B
5.19.1	C